

Reciprocal Five-Term Relations among p^j -th Roots of Unity in Characteristic Two: Reductions, Obstructions, and the Wieferich Plateau

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Abstract

Let $p > 5$ be prime, let $j \geq 2$, and let $\theta \in \overline{\mathbb{F}}_2$ have exact order p^j . We study the equation

$$1 + \theta^a + \theta^{-a} + \theta^b + \theta^{-b} = 0,$$

whose only solutions are conjectured to have $p \mid a$ and $p \mid b$. This is equivalent to the stabilization $d(p^k - 1) = d(p - 1)$ of the nullity of the bounded *Lights Out* grid. This article consolidates the algebraic, cyclotomic, local, additive, coding-theoretic, dynamical, elliptic, and symplectic attacks on that conjecture, including the points at which each method becomes sharp but inconclusive.

The equation is encoded by the reciprocal quartic $Q_Y(T) = T^4 + T^3 + YT^2 + T + 1$ and by a shifted-Dickson sequence $C_n(S)$ whose repeated roots are exactly the quartics dividing $T^n + 1$. We prove that every repeated root of C_n has multiplicity exactly two, with $C_n^{[2]}(Y) = Y^{-2}$, and obtain an exact bridge to Sutner's nullity formula. An integer lift factors over the maximal real cyclotomic field and turns stabilization into the simultaneous oddness of an internal and a mixed collision norm. Arithmetic dynamics gives the same off-plateau descent through the degree-two covering $u \mapsto u + u^{-1}$ and identifies irreducible factors with doubling orbits of collision pairs. The equation also becomes a translated coordinate intersection on an elliptic curve and the trace-one locus of a regular semisimple element of Sp_4 ; both reformulations clarify the geometry but return to the original incidence problem on the plateau.

If $c_p = v_p(2^{\mathrm{ord}_p(2)} - 1)$, every solution has $\mathrm{ord}(\theta^a), \mathrm{ord}(\theta^b) \leq p^{c_p}$, so only $2 \leq j \leq c_p$ remains. The whole theorem is therefore equivalent to one endpoint assertion at order p^{c_p} . Characteristic-zero vanishing sums, cyclotomic units, ray classes, power residues, modular vanishing-sum theorems, cyclic-code bounds, and maximal-torus classification all stop at that same endpoint. The semi-local residue module is explicitly $\mathbb{Z}[\tilde{G}]/(\sigma_2 - 2)$, and its Frobenius, sign, distribution, and norm relations leave the primitive orbit values freely parameterized; explicit phantom systems show that these relations, even together with the known trace constraints, do not force avoidance.

Genuinely additive input does prove new cases. The Artin-Schreier law $\mathrm{Tr}(\bar{\lambda}_x^{-1}) = \delta$ and the global law $\mathrm{Tr}(\theta^x) + \mathrm{Tr}(\theta^y) = \mathrm{ord}_{p^j}(2) \bmod 2$ exclude several families, including $d(3510) = d(72) = d(88) = d(232) = 0$. At the opposite extreme, the Mersenne and Fermat cases admit exact binary Kloosterman formulas. The statement is false in other residue characteristics, so any proof must use characteristic-two additive structure. Exact searches verify the endpoint for both known base-2 Wieferich primes, 1093 and 3511, and a complete resultant computation settles $p^j = 49$. No proof is presently known for a hypothetical new Wieferich endpoint.

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1 The problem

Let $F_n(x) \in \mathbb{F}_2[x]$ be the Fibonacci polynomials

$$F_0(x) = 0, \quad F_1(x) = 1, \quad F_n(x) = xF_{n-1}(x) + F_{n-2}(x),$$

and let $d(n)$ be the nullity over $\mathbb{F}_2$ of the $n \times n$ bounded *Lights Out* grid, so that

$$d(n) = \deg \gcd(F_{n+1}(x), F_{n+1}(x+1)) \tag{1}$$

by Sutner's formula [Sut00]. For odd $N > 1$ the distinct roots of F_N are

$$\bigcup_{\substack{M|N \\ M>1}} R_M, \quad R_M = \{\zeta + \zeta^{-1} : \text{ord}(\zeta) = M\}, \tag{2}$$

and every root has multiplicity two [HMP04].

A companion manuscript at `tex/finite_fields/finite_fields.tex` proves the *prime stabilization theorem*: with

$$c = c_p = v_p\left(2^{\text{ord}_p(2)} - 1\right), \tag{3}$$

one has $d(p^k - 1) = d(p^c - 1)$ for all $k \geq c$. Since $c = 1$ unless p is a Wieferich prime, this settles $d(p^k - 1) = d(p - 1)$ for every non-Wieferich prime, and exact computations settle the two known Wieferich primes 1093 and 3511. The residual obstruction is a single, purely finite-field statement.

Problem 1.1. *Let $p > 5$ be prime, let $j \geq 2$, and let $\theta \in \overline{\mathbb{F}_2}$ have exact order p^j . Suppose that*

$$1 + \theta^a + \theta^{-a} + \theta^b + \theta^{-b} = 0. \tag{4}$$

Must $p \mid a$ and $p \mid b$?

The hypothesis $p > 5$ is genuinely needed: for $p = 5$ and $j = 1$ the relation $1 + \theta + \theta^4 + \theta^2 + \theta^3 = 0$ holds in $\mathbb{F}_{16}$. Section 7 explains why 5 is the only obstruction of this kind.

Throughout, $u = \theta^a$ and $v = \theta^b$, and we write

$$S_N = \{(a, b) \in (\mathbb{Z}/N\mathbb{Z})^2 : 1 + \theta^a + \theta^{-a} + \theta^b + \theta^{-b} = 0\}$$

for θ of exact order N .

The organization follows the successive changes of language used in the attack. Section 2 develops the shifted-Dickson, reciprocal-quartic, elliptic, and nullity dictionaries. Section 3 proves descent off the Wieferich plateau and states the single endpoint lemma equivalent to the full theorem. Sections 4 and 5 give the integer, real-cyclotomic, and arithmetic-dynamical forms of the same obstruction. Section 6 records the symplectic reformulation and explains why maximal-torus and character theory do not decide the trace-one incidence. Sections 7–8 treat characteristic zero and the ratios eliminated by cyclotomic units. Sections 9 and 10 develop the local circular-unit, semi-local module, ray-class, power-residue, and additive trace attacks. Sections 11 and 12 explain the exact limitations of modular vanishing sums and cyclic/Zetterberg codes. Sections 13 and 14 record the counterexamples in other characteristics and the size heuristic. Section 15 lists the exact certificates, and Section 16 collects what is proved, what failed, and what remains.

1.1 Elementary normalizations

Lemma 1.2. *Let $N > 1$ be odd with $3 \nmid N$, let $\theta \in \overline{\mathbb{F}}_2$ have exact order N , and let $(a, b) \in S_N$. Then $a \not\equiv 0$, $b \not\equiv 0$ and $b \not\equiv \pm a$ modulo N . In particular the five roots of unity $1, \theta^{\pm a}, \theta^{\pm b}$ are pairwise distinct.*

Proof. If $a \equiv 0$ then $\theta^a + \theta^{-a} = 1 + 1 = 0$, so $\theta^b + \theta^{-b} = 1$ and θ^b is a root of $T^2 + T + 1$, hence of exact order 3. Then $3 \mid N$, a contradiction. The case $b \equiv 0$ is symmetric. If $b \equiv \pm a$ then $\theta^b + \theta^{-b} = \theta^a + \theta^{-a}$ and the left side of (4) equals 1. ■

Lemma 1.3. *S_N is stable under $(a, b) \mapsto (2a, 2b)$, under $(a, b) \mapsto (\pm a, \pm b)$ and under $(a, b) \mapsto (b, a)$. If θ is replaced by θ^t with $\gcd(t, N) = 1$ then S_N is replaced by $t^{-1}S_N$. In particular whether S_N is empty does not depend on the choice of θ .*

Proof. The first claim is the Frobenius $x \mapsto x^2$ applied to (4); the second and third are the substitutions $a \mapsto -a$ and $a \leftrightarrow b$, which permute the summands. The last claim is immediate from $1 + (\theta^t)^a + \dots = 1 + \theta^{ta} + \dots$. ■

1.2 Why the problem is exactly the residual obstruction

Lemma 1.4. *Let $N > 1$ be odd with $3 \nmid N$ and let $\theta \in \overline{\mathbb{F}}_2$ have exact order N . Then*

$$d(N - 1) = \frac{1}{2} |S_N|.$$

Proof. By (2) the roots of F_N are exactly the elements $u + u^{-1}$ with $u \in \mu_N \setminus \{1\}$, each of multiplicity two. A common root of $F_N(x)$ and $F_N(x + 1)$ is an α with $\alpha = u + u^{-1}$ and $\alpha + 1 = v + v^{-1}$ for some $u, v \in \mu_N \setminus \{1\}$, i.e. with $1 + u + u^{-1} + v + v^{-1} = 0$. Writing $u = \theta^a$ and $v = \theta^b$ identifies the set of such α with S_N modulo the action of $\{\pm 1\} \times \{\pm 1\}$ on (a, b) ; the action is free because $a \equiv -a$ would force $a \equiv 0$, which Lemma 1.2 excludes. Hence the number of common roots is $|S_N|/4$, and (1) together with multiplicity two gives $d(N - 1) = 2 \cdot |S_N|/4$. ■

Corollary 1.5. *If Problem 1.1 has an affirmative answer for a prime $p > 5$ and all $j \geq 2$, then $d(p^k - 1) = d(p - 1)$ for all $k \geq 1$.*

Proof. Let $k \geq 2$ and let θ have exact order p^k . Any element of S_{p^k} has $p \mid a$ and $p \mid b$, so $\theta^a, \theta^b \in \mu_{p^{k-1}}$ and the same relation is an element of $S_{p^{k-1}}$ after replacing θ by θ^p . Iterating, $|S_{p^k}| = |S_p|$, and Lemma 1.4 gives $d(p^k - 1) = d(p - 1)$. ■

Lemma 1.4 was verified numerically against the independent grid-nullity code in `code/kernel_size.py`: for $p = 5, 17, 31$ the counts $|S_p| = 8, 16, 40$ match $d(4) = 4$, $d(16) = 8$, $d(30) = 20$, and for $p = 7, 11, 41, 43, 73$ the empty solution sets match $d(6) = d(10) = d(40) = d(42) = d(72) = 0$.

1.3 The reciprocal quartic

Set

$$Q_Y(T) = T^4 + T^3 + YT^2 + T + 1 \in \overline{\mathbb{F}}_2[Y][T],$$

a monic reciprocal quartic whose roots have product 1.

Lemma 1.6. *Let $\alpha \in \overline{\mathbb{F}}_2$ and $Y = \alpha^2 + \alpha$. Then*

$$Q_Y(T) = (T^2 + \alpha T + 1)(T^2 + (\alpha + 1)T + 1).$$

Consequently, for $u, v \in \overline{\mathbb{F}}_2^\times$ the following are equivalent.

1. $1 + u + u^{-1} + v + v^{-1} = 0$.
2. u, u^{-1}, v, v^{-1} are exactly the roots of Q_Y for $Y = \alpha^2 + \alpha$ with $\alpha = u + u^{-1}$.

Moreover Problem 1.1 has a negative answer for the pair (p, j) if and only if there is $Y \in \overline{\mathbb{F}}_2$ such that $Q_Y(T) \mid T^{p^j} + 1$ and at least one root of Q_Y has exact order p^j .

Proof. Expanding the product gives $T^4 + (\alpha + \alpha + 1)T^3 + (1 + \alpha(\alpha + 1) + 1)T^2 + (\alpha + \alpha + 1)T + 1$, which is $Q_{\alpha^2 + \alpha}(T)$. The quadratic $T^2 + \alpha T + 1$ has root set $\{u, u^{-1}\}$ precisely when $u + u^{-1} = \alpha$; likewise $T^2 + (\alpha + 1)T + 1$ has root set $\{v, v^{-1}\}$ precisely when $v + v^{-1} = \alpha + 1$, which is statement (1). For the last assertion, $T^{p^j} + 1$ has exactly the elements of μ_{p^j} as roots, each simple, so $Q_Y \mid T^{p^j} + 1$ says precisely that $u, v \in \mu_{p^j}$; and a counterexample to Problem 1.1 is exactly a relation in which θ^a or θ^b has exact order p^j . ■

The next lemma identifies the resultant that appears in Lemma 1.6 with a Fibonacci-polynomial expression, and is the bridge between the quartic picture and (1). Let D_N denote the Dickson polynomial of the first kind [LN97, Ch. 7], determined by $D_N(t + t^{-1}) = t^N + t^{-N}$.

Lemma 1.7. *Over $\mathbb{F}_2$ one has $D_N(s) = s F_N(s)$ for every $N \geq 0$. Consequently, for odd N ,*

$$\text{Res}_T(Q_Y(T), T^N + 1) = Y \cdot \text{Res}_s(s^2 + s + Y, F_N(s)),$$

and for $Y = \alpha^2 + \alpha$ with $\alpha \neq 0, 1$,

$$\text{Res}_s(s^2 + s + Y, F_N(s)) = F_N(\alpha)F_N(\alpha + 1).$$

Proof. Both D_N and sF_N satisfy the recursion $X_N = sX_{N-1} + X_{N-2}$ over $\mathbb{F}_2$; the Dickson recursion is $D_N = sD_{N-1} - D_{N-2}$, and $-1 = 1$. The initial values agree, since $D_0 = 2 = 0 = s \cdot F_0$ and $D_1 = s = s \cdot F_1$.

For the resultant, let $t_1, \dots, t_4$ be the roots of Q_Y , paired as $\{t, t^{-1}\}$ by Lemma 1.6. In characteristic two, $(t^N + 1)(t^{-N} + 1) = t^N + t^{-N} = D_N(t + t^{-1})$, so

$$\text{Res}_T(Q_Y, T^N + 1) = \prod_{i=1}^4 (t_i^N + 1) = D_N(\alpha)D_N(\alpha + 1) = \alpha(\alpha + 1)F_N(\alpha)F_N(\alpha + 1),$$

and $\alpha(\alpha + 1) = \alpha^2 + \alpha = Y$. The roots of $s^2 + s + Y$ are α and $\alpha + 1$, which gives the second displayed identity. $\blacksquare$

2 Shifted-Dickson, nullity, and elliptic reformulations

The reciprocal quartic can be packaged into one explicit polynomial sequence. This is useful for two reasons: it records the repeated-root locus with its exact multiplicity, and it turns the desired stabilization into ordinary square-freeness and coprimality statements.

Definition 2.1. *Since $D_n(X) + D_n(X + 1)$ is invariant under $X \mapsto X + 1$, there is a unique polynomial $C_n(S) \in \mathbb{F}_2[S]$ such that*

$$C_n(X^2 + X) = D_n(X) + D_n(X + 1). \quad (5)$$

Proposition 2.2 (Power sums and generating function). *The polynomial $C_n(S)$ is the n -th Newton power sum of the four roots of $Q_S(T)$:*

$$C_n(S) = \sum_{Q_S(\rho)=0} \rho^n.$$

Moreover

$$\sum_{n \geq 0} C_n(S) t^n = \frac{t(1+t)^2}{Q_S(t)} = \frac{t + t^3}{1 + t + St^2 + t^3 + t^4}. \quad (6)$$

Hence

$$C_{n+4} = C_{n+3} + SC_{n+2} + C_{n+1} + C_n, \quad (C_0, C_1, C_2, C_3) = (0, 1, 1, S).$$

Proof. Write $S = X^2 + X$. By Lemma 1.6, the roots of Q_S are $\zeta, \zeta^{-1}, \eta, \eta^{-1}$ with $\zeta + \zeta^{-1} = X$ and $\eta + \eta^{-1} = X + 1$. Their n -th power sum is therefore $D_n(X) + D_n(X + 1) = C_n(S)$. If $g(t) = Q_S(t)$, reciprocity gives $\prod_{\rho} (1 - \rho t) = g(t)$, and

$$-\frac{tg'(t)}{g(t)} = \sum_{\rho} \frac{\rho t}{1 - \rho t} = \sum_{n \geq 1} \left(\sum_{\rho} \rho^n \right) t^n.$$

In characteristic two, $g'(t) = 1 + t^2 = (1 + t)^2$ and $C_0 = 4 = 0$. This proves (6); clearing the denominator gives the recurrence. $\blacksquare$

Proposition 2.3 (Elementary identities). *For all $m, n \geq 1$:*

$$C_{2n} = C_n^2, \quad m \mid n \implies C_m \mid C_n.$$

Proof. The first identity is Frobenius applied to the power sum. For the second write $n = mk$ and use Dickson composition:

$$D_n(X) + D_n(X + 1) = D_k(D_m(X)) + D_k(D_m(X + 1)).$$

The difference $A + B$ divides $D_k(A) + D_k(B)$ in characteristic two, so the right side is divisible by $D_m(X) + D_m(X + 1)$. The substitution $\mathbb{F}_2[S] \hookrightarrow \mathbb{F}_2[X]$, $S \mapsto X^2 + X$, is injective. $\blacksquare$

For $f(S) = \sum_i a_i S^i$, write $f^{[r]} = \sum_i a_i \binom{i}{r} S^{i-r}$ for its r -th Hasse derivative.

Proposition 2.4 (Hasse-derivative generating function). *For every $r \geq 0$,*

$$\sum_{n \geq 0} C_n^{[r]}(S) t^n = \frac{t^{2r+1}(1+t)^2}{Q_S(t)^{r+1}}. \quad (7)$$

Proof. The polynomial $Q_S(t) = t^2S + (1+t+t^3+t^4)$ is affine in S . The Hasse expansion of its reciprocal gives $\partial_S^{[r]}Q_S(t)^{-1} = t^{2r}Q_S(t)^{-r-1}$. Apply this to (6). ■

For odd n , the polynomial F_n is the square of a square-free polynomial [HMP04, Prop. 2.11]; write

$$F_n(X) = H_n(X)^2.$$

Then $D_n(X) = XH_n(X)^2$ and

$$C_n(X^2 + X) = XH_n(X)^2 + (X+1)H_n(X+1)^2. \quad (8)$$

Lemma 2.5 (Repeated roots and dividing quartics). *Let n be odd. A value $Y \in \overline{\mathbb{F}_2}$ is a repeated root of C_n if and only if*

$$Y = X_0^2 + X_0$$

for a common root X_0 of $F_n(X)$ and $F_n(X+1)$. Equivalently, $Y \neq 0$ and $Q_Y(T) \mid T^n + 1$.

Proof. Differentiating (8) gives

$$\frac{d}{dX}C_n(X^2 + X) = H_n(X)^2 + H_n(X+1)^2.$$

At a repeated root X_0 , the two values $H_n(X_0)$ and $H_n(X_0+1)$ are equal. Substitution in (8) then shows that their common value is zero. The converse is immediate. The map $X \mapsto X^2 + X$ identifies the pair $\{X_0, X_0+1\}$, and Lemma 1.6 together with the standard root parametrization of F_n gives the quartic criterion. Since $F_n(0) = 1$ for odd n , neither $X_0 = 0$ nor $X_0 = 1$ occurs, so $Y \neq 0$. ■

Theorem 2.6 (Multiplicity Two). *If n is odd and Y is a repeated root of C_n , then*

$$C_n^{[2]}(Y) = Y^{-2} \neq 0. \quad (9)$$

Thus every repeated root has multiplicity exactly two, and

$$C_n = A_n B_n^2, \quad \gcd(C_n, C'_n) = B_n^2, \quad (10)$$

where A_n and B_n are square-free and coprime and the roots of B_n are precisely the values in Lemma 2.5.

Proof. Let $Y = X_0^2 + X_0$ be repeated, and write the two reciprocal root pairs of Q_Y as $\{\zeta, \zeta^{-1}\}$ and $\{\eta, \eta^{-1}\}$, where $X_0 = \zeta + \zeta^{-1}$ and $X_0 + 1 = \eta + \eta^{-1}$. All four roots are distinct n -th roots of unity. Put $m = (n-1)/2$. If $X = t + t^{-1}$, then

$$H_n(X) = \frac{t^n + 1}{t^m(t+1)}.$$

Differentiating at $t = \zeta$ and using $dX/dt = (t+1)^2/t^2$ gives

$$H'_n(X_0)^2 = \frac{\zeta^3}{(\zeta+1)^6},$$

and the analogous identity holds with η and $X_0 + 1$. The coefficient of $(X - X_0)^2$ in (8) is therefore

$$X_0 H'_n(X_0)^2 + (X_0 + 1) H'_n(X_0 + 1)^2 = \frac{\zeta^2}{(\zeta+1)^4} \frac{\eta^2}{(\eta+1)^4}.$$

Since $\zeta/(\zeta+1)^2 = X_0^{-1}$ and $\eta/(\eta+1)^2 = (X_0 + 1)^{-1}$, this is

$$\left(\frac{1}{X_0} + \frac{1}{X_0 + 1} \right)^2 = (X_0^2 + X_0)^{-2} = Y^{-2}.$$

The substitution $S = X^2 + X$ has derivative 1, so this coefficient is exactly $C_n^{[2]}(Y)$. The factorization follows. ■

Theorem 2.7 (Bridge to Sutner's nullity). *For odd n ,*

$$\gcd(F_n(X), F_n(X+1)) = \gcd(C_n, C'_n)(X^2 + X). \quad (11)$$

In particular

$$d(n-1) = 4 \deg B_n = 2 \deg \gcd(C_n, C'_n).$$

Proof. With $F_n = H_n^2$, the square-free polynomial $\gcd(H_n(X), H_n(X+1))$ has roots in translation pairs $\{X_0, X_0+1\}$. Lemma 2.5 identifies the polynomial obtained after quotienting those pairs as B_n , so

$$\gcd(H_n(X), H_n(X+1)) = B_n(X^2 + X).$$

Square and use (10). The degree statement follows from Sutner's formula (1). ■

2.1 Product coordinates and an elliptic translation

There is also a geometric normalization that retains both multiplicative coordinates. Put

$$z = uv, \quad r = u/v.$$

Then, in characteristic two,

$$1 + u + u^{-1} + v + v^{-1} = 0 \iff (z + z^{-1})(r + r^{-1}) = 1. \quad (12)$$

Indeed the right side is the square of $u + u^{-1} + v + v^{-1}$.

Proposition 2.8 (Elliptic translation model). *Let*

$$E: Y^2 + XY = X^3 + X, \quad T = (1, 0) \in E(\mathbb{F}_2).$$

The point T has order four. Away from the degenerate values $z, r \in \{0, 1\}$, the curve in (12) is isomorphic to the open subset of E under

$$P = (z, (r+1)(z+1)^2).$$

Under this correspondence

$$X(P) = z, \quad X(P+T) = r.$$

Consequently a five-term relation of conductor n is a point P for which both $X(P)$ and $X(P+T)$ lie in μ_n .

Proof. Set $y = (r+1)(z+1)^2$. After clearing denominators, (12) is

$$z^2 r^2 + zr + z^2 + r^2 + 1 = 0,$$

which is equivalent to $y^2 + zy = z^3 + z$. Thus $(z, y) \in E$. The addition law on the binary Weierstrass model gives

$$X(P+T) = 1 + \frac{Y(P)}{(X(P)+1)^2} = r.$$

Conversely, the curve equation implies

$$(X + X^{-1})(X(P+T) + X(P+T)^{-1}) = 1.$$

A direct doubling calculation gives $2T = (0, 0)$ and $2(0, 0) = O$. ■

Remark 2.9 (What the elliptic model does and does not give). *The model explains the genus-one geometry behind the equation and relates the two multiplicative coordinates by translation by a rational four-torsion point. It does not turn the problem into an ordinary torsion question on E : the condition is that two X -coordinates lie in the subgroup $\mu_{p^j} \subset \mathbb{F}_2^\times$, not that P belongs to a specified subgroup of E . Isogenies, complex multiplication, and point-count estimates therefore return to the same subgroup-incidence problem. This is why the elliptic attack gives exact formulas in full-torus cases but does not resolve a Wieferich plateau.*

3 The degree argument and the Wieferich plateau

For odd $M > 1$ put

$$\rho_2(M) = \min\{r > 0 : 2^r \equiv \pm 1 \pmod{M}\},$$

so that $[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(M)$ for $\alpha \in R_M$; this is the field-extension-size lemma of the companion manuscript at `tex/finite_fields/finite_fields.tex`.

Lemma 3.1. *Let p be an odd prime and $c = c_p$ as in (3). Then for every $i \geq 1$,*

$$\text{ord}_{p^i}(2) = \text{ord}_p(2) p^{\max(0, i-c)}, \quad \rho_2(p^i) = \rho_2(p) p^{\max(0, i-c)}.$$

Proof. The first identity is the lifting-the-exponent computation carried out in `tex/finite_fields/finite_fields.tex`. For the second, $\rho_2(M) = \text{ord}_M(2)$ or $\text{ord}_M(2)/2$ according as $-1 \notin \langle 2 \rangle$ or $-1 \in \langle 2 \rangle$ in $(\mathbb{Z}/M\mathbb{Z})^\times$. The group $(\mathbb{Z}/p^i\mathbb{Z})^\times$ is cyclic and contains a unique element of order two, namely -1 ; hence $-1 \in \langle 2 \rangle$ if and only if $\text{ord}_{p^i}(2)$ is even, i.e. if and only if $\text{ord}_p(2)$ is even, because $p^{\max(0, i-c)}$ is odd. The criterion is therefore independent of i , and the second identity follows from the first. ■

Definition 3.2. *The base-2 plateau of p is the set of exponents $1 \leq i \leq c_p$. Equivalently, i lies on the plateau if and only if $\mu_{p^i} \subseteq \mathbb{F}_{2^{f_1}}$ with $f_1 = \text{ord}_p(2)$, if and only if 2 splits completely in $\mathbb{Q}(\zeta_{p^i})/\mathbb{Q}(\zeta_p)$. The plateau has length ≥ 2 exactly when p is a Wieferich prime.*

Proposition 3.3 (Frobenius-orbit restriction). *Suppose (4) holds, $\gcd(a, p) = 1$, and $b \equiv \pm 2^k a \pmod{p^j}$. Put $d = \rho_2(p^j)$ and reduce k modulo d to $0 \leq k < d$. Then d is even and $k = d/2$. In particular, when $\rho_2(p^j)$ is odd, a counterexample cannot pair two trace values in the same Frobenius orbit.*

Proof. Set $\alpha = \theta^a + \theta^{-a}$. Since θ^a has exact order p^j , one has $[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = d$. The congruence on b and (4) give

$$\alpha^{2^k} = \theta^b + \theta^{-b} = \alpha + 1.$$

Applying the same Frobenius power once more yields $\alpha^{2^{2k}} = (\alpha + 1)^{2^k} = \alpha$, so $d \mid 2k$. One cannot have $d \mid k$, since that would give $\alpha^{2^k} = \alpha \neq \alpha + 1$. With $0 < k < d$, the divisibility $d \mid 2k$ therefore forces d even and $k = d/2$. ■

Proposition 3.4 (Orbit divisibility). *Let $\Delta_j = |S_{p^j}| - |S_{p^{j-1}}|$, where the lower solution set is embedded by $(a, b) \mapsto (pa, pb)$, and put $\rho = \rho_2(p^j)$. Then*

$$\Delta_j \in \begin{cases} 8\rho\mathbb{Z}, & \rho \text{ odd}, \\ 4\rho\mathbb{Z}, & \rho \text{ even}. \end{cases}$$

Thus any nonzero new stratum contains at least 8ρ ordered solutions when ρ is odd and at least 4ρ when ρ is even.

Proof. On solution pairs $(u, v) = (\theta^a, \theta^b)$, independent inversion of the two coordinates and transposition generate a dihedral group of order 8. Its action is free for $p > 3$: a fixed inversion forces a coordinate to be 1, while a fixed signed transposition forces $v = u^{\pm 1}$ and then $(u + u^{-1})^2 = 1$, so u has order 3.

Frobenius acts by $(u, v) \mapsto (u^2, v^2)$. If $f = \text{ord}_{p^j}(2)$ is odd, then $f = \rho$ and its cyclic group meets the dihedral group trivially. If f is even, then $f = 2\rho$ and Frobenius to the power ρ is simultaneous inversion, so the combined effective group still has order 8ρ .

A nontrivial stabilizer in this effective group must involve a signed transposition. With one coordinate primitive, such a stabilizer gives $2^{2k} \equiv -1 \pmod{p^j}$; the $+1$ alternative again forces $v = u^{\pm 1}$. Hence it can occur only when ρ is even, with $k \equiv \rho/2 \pmod{\rho}$, and then its order is exactly 2. The new solutions therefore split into orbits of size 8ρ when ρ is odd and of size 8ρ or 4ρ when ρ is even. ■

Theorem 3.5. *Let $p > 3$ be prime, let $\theta \in \overline{\mathbb{F}}_2$ have exact order p^j , and suppose (4) holds. Then*

$$\text{ord}(\theta^a) \leq p^c, \quad \text{ord}(\theta^b) \leq p^c.$$

In particular p^{j-c} divides both a and b whenever $j > c$.

Proof. By Lemma 1.2 the orders $p^{i_1} = \text{ord}(\theta^a)$ and $p^{i_2} = \text{ord}(\theta^b)$ satisfy $i_1, i_2 \geq 1$. Put $\alpha = \theta^a + \theta^{-a}$, so $\alpha \in R_{p^{i_1}}$ and $\alpha + 1 \in R_{p^{i_2}}$ by (4). Since $\mathbb{F}_2(\alpha) = \mathbb{F}_2(\alpha + 1)$, we get $\rho_2(p^{i_1}) = \rho_2(p^{i_2})$, whence $\max(0, i_1 - c) = \max(0, i_2 - c)$ by Lemma 3.1. Therefore either $i_1, i_2 \leq c$, and we are done, or $i_1 = i_2 =: i > c$.

Assume $i > c$. Both θ^a and θ^b have exact order p^i , so writing $\zeta = \theta^a$ we have $\theta^b = \zeta^s$ for some s with $\gcd(s, p) = 1$, and ζ is a root of

$$P(x) = 1 + x + x^{p^i-1} + x^s + x^{p^i-s} \in \mathbb{F}_2[x]. \quad (13)$$

Let $m = p^{i-c} > 1$. Then ζ^m has exact order p^c , and its minimal polynomial g over $\mathbb{F}_2$ has degree $\text{ord}_{p^c}(2) = \text{ord}_p(2)$ by Lemma 3.1. Hence

$$\deg g(x^m) = \text{ord}_p(2) p^{i-c} = \text{ord}_{p^i}(2),$$

which is the degree of the minimal polynomial of ζ ; as ζ is a root of $g(x^m)$, that polynomial is the minimal polynomial of ζ , and therefore $g(x^m) \mid P(x)$.

Decompose $P(x) = \sum_{r=0}^{m-1} x^r P_r(x^m)$, which is possible in exactly one way. Writing $P = g(x^m)H$ and $H = \sum_r x^r H_r(x^m)$ and comparing the unique decompositions gives $P_r = gH_r$ for every r ; in particular $g \mid P_0$. The five exponents in (13) reduce modulo m to 0, 1, -1 , s , $-s$. Since m is an odd prime power exceeding 1 and $\gcd(s, p) = 1$, none of 1, -1 , s , $-s$ is $\equiv 0 \pmod{m}$. Hence $P_0 = 1$ and $g \nmid 1$, contradicting $\deg g \geq 1$. ■

Corollary 3.6. *Problem 1.1 has an affirmative answer for every non-Wieferich prime $p > 3$ and every $j \geq 2$, and more generally whenever $j > c_p$. Consequently a counterexample requires a Wieferich prime p with $c_p \geq 2$ and an exponent j with $2 \leq j \leq c_p$; that is, the problem is open exactly on the base-2 Wieferich plateau.*

Proof. If p is not Wieferich then $c_p = 1 < 2 \leq j$, so Theorem 3.5 gives $p^{j-1} \mid a, b$, and in particular $p \mid a, b$. If $j > c_p$ the same argument applies verbatim. A counterexample therefore has $j \leq c_p$, and $j \geq 2$ forces $c_p \geq 2$. ■

Corollary 3.7. *Problem 1.1 has an affirmative answer for every prime $5 < p < 2^{64}$ and every $j \geq 2$. Hence $d(p^k - 1) = d(p - 1)$ for every such p and every k .*

Proof. The only Wieferich primes below 2^{64} are 1093 and 3511 [CDP97, KR05, DK11, Pri26], and both have $c_p = 2$. Corollary 3.6 handles every other p , and handles $j \geq 3$ for $p \in \{1093, 3511\}$. The remaining case $j = 2$ for $p \in \{1093, 3511\}$ is settled by the direct searches of Section 15, which find no solutions; alternatively it follows from the exact polynomial gcd evaluations $d(1093^2 - 1) = d(3511^2 - 1) = 0$ recorded in `tex/finite_fields/finite_fields.tex` together with Lemma 1.4. Corollary 1.5 converts this into the statement about d . ■

Theorem 3.5 is a small sharpening of the argument in the companion manuscript, which treats only the case of equal orders. The point of stating it in this form is Corollary 3.6: the plateau is not merely where the known proof stops, it is the exact locus of the residual problem. Section 13 shows that on the analogous plateau in other residue characteristics the statement can genuinely fail.

3.1 The single endpoint statement

The preceding reduction can be sharpened from a finite list of plateau levels to one statement at the top of the plateau.

Theorem 3.8 (Endpoint equivalence). *Fix a prime $p > 5$ with $c = c_p \geq 2$. The following are equivalent.*

1. *Problem 1.1 has an affirmative answer for every $j \geq 2$.*
2. *Whenever Θ has exact order p^c and*

$$1 + \Theta^a + \Theta^{-a} + \Theta^b + \Theta^{-b} = 0,$$

one has $p^{c-1} \mid a$ and $p^{c-1} \mid b$.

3. $B_{p^c} = B_p$.
4. *Writing $F_n = H_n^2$ as in Section 2,*

$$\gcd\left(\frac{H_{p^c}(X)}{H_p(X)}, H_{p^c}(X+1)\right) = 1. \quad (14)$$

Proof. By Theorem 3.5, levels above c introduce no new solution. Thus (1) implies (2). Conversely suppose (2), and let $2 \leq j \leq c$. Choose Θ of order p^c and put $\theta = \Theta^{p^{c-j}}$. A relation at level p^j becomes one at level p^c with exponents $p^{c-j}a, p^{c-j}b$. Statement (2) then gives $p^{j-1} \mid a, b$, hence in particular $p \mid a, b$. This proves (1).

Lemma 2.5 identifies the roots of B_{p^k} with the common-root pairs whose two multiplicative orders divide p^k . Therefore $B_{p^c} = B_p$ says exactly that every endpoint relation has both orders dividing p , which is (2). Finally, the quotient H_{p^c}/H_p records the roots whose order is new above level p ; its coprimality with the translate of H_{p^c} is exactly the absence of a common-root pair with at least one new coordinate. This is (3). ■

Definition 3.9. *For $j \geq 1$ define the shifted-Dickson primitive quotient*

$$\Pi_{p^j}(S) = \frac{C_{p^j}(S)}{C_{p^{j-1}}(S)}, \quad C_{p^0} = C_1 = 1.$$

The letter Π is used to avoid confusion with the cyclotomic polynomial Φ_{p^j} .

Proposition 3.10 (Polynomial stabilization criterion). *For every $j \geq 1$ the following are equivalent:*

1. $B_{p^j} = B_{p^{j-1}}$;
2. $\gcd(C_{p^j}, C'_{p^j}) = \gcd(C_{p^{j-1}}, C'_{p^{j-1}})$;
3. Π_{p^j} is square-free and coprime to $C_{p^{j-1}}$.

Moreover $B_{p^j} = B_{p^c}$ for every $j \geq c$, so the full theorem is equivalent to $B_{p^c} = B_p$.

Proof. The first two statements are equivalent by (10). Because every repeated root of every C_n has exact multiplicity two, a new repeated root at level p^j either appears twice in Π_{p^j} or appears once in both Π_{p^j} and $C_{p^{j-1}}$; these are exactly the two failures in (3). The final statement is Theorem 3.5 translated through Lemma 2.5. ■

4 Integer lifts and cyclotomic collision norms

The shifted-Dickson target has a useful integral lift. It does not remove the endpoint obstruction, but it isolates it as the parity of two explicit integer norms and makes clear that both a primitive–primitive and a primitive–lower collision must be excluded.

Proposition 4.1 (Integer lift). *For odd n there is a monic polynomial $\tilde{C}_n \in \mathbb{Z}[Y]$ such that*

$$D_n(X) + D_n(X+1) = (2X+1)\tilde{C}_n(X^2+X), \quad \tilde{C}_n \equiv C_n \pmod{2}. \quad (15)$$

Proof. The involution $\iota(X) = -1 - X$ fixes $Y = X^2 + X$. For odd n , the integer Dickson polynomial D_n is odd, so $D_n(X) + D_n(X+1)$ is anti-invariant under ι . Write an arbitrary element of $\mathbb{Z}[X]$ uniquely as $A(Y) + XB(Y)$. Anti-invariance forces $B = 2A$, and hence the polynomial is $(2X+1)A(Y)$. Its leading term shows that $A = \tilde{C}_n$ is monic. Reducing modulo two gives (5). ■

For odd k , put

$$L_k[A, B] = \frac{D_k(A) + D_k(B)}{A+B} \in \mathbb{Z}[A, B].$$

Proposition 4.2 (Composition and telescoping). *If m and k are odd, then*

$$\frac{\tilde{C}_{mk}}{\tilde{C}_m} = L_k[D_m(X), D_m(X+1)] \in \mathbb{Z}[Y].$$

In particular $\tilde{C}_m \mid \tilde{C}_n$ whenever $m \mid n$, and

$$\tilde{\Pi}_{p^j}(Y) := \frac{\tilde{C}_{p^j}(Y)}{\tilde{C}_{p^{j-1}}(Y)} = L_p[D_{p^{j-1}}(X), D_{p^{j-1}}(X+1)]. \quad (16)$$

Proof. Dickson composition gives

$$D_{mk}(X) + D_{mk}(X+1) = D_k(A) + D_k(B) = (A+B)L_k[A, B],$$

with $A = D_m(X)$ and $B = D_m(X+1)$. The quotient is invariant under $X \mapsto -1 - X$, hence lies in $\mathbb{Z}[Y]$. Cancel the common factor $2X+1$ from (15). ■

Theorem 4.3 (Discriminant/resultant certificate). *At level p^j , the equality $B_{p^j} = B_{p^{j-1}}$ holds if and only if both integers*

$$\text{disc}(\tilde{\Pi}_{p^j}) \quad \text{and} \quad \text{Res}(\tilde{\Pi}_{p^j}, \tilde{C}_{p^{j-1}})$$

are odd.

Proof. Reduction modulo two commutes with discriminants and resultants. Thus the first integer is odd exactly when Π_{p^j} is square-free, and the second is odd exactly when Π_{p^j} is coprime to $C_{p^{j-1}}$. Apply Proposition 3.10. ■

4.1 Factorization over the maximal real cyclotomic field

Let

$$\lambda_k = \zeta_n^k + \zeta_n^{-k}, \quad 1 \leq k \leq \frac{n-1}{2},$$

and let ψ_m be the minimal polynomial of $\zeta_m + \zeta_m^{-1}$. Put

$$\mathcal{R}_n(\lambda) = \prod_{\substack{d \mid n \\ d > 1}} \psi_d(\lambda).$$

Proposition 4.4 (Real-cyclotomic factorization). *For odd n ,*

$$\tilde{C}_n(Y) = \prod_{k=1}^{(n-1)/2} ((\lambda_k + 2)Y + \lambda_k^2 - 3) = \text{Res}_\lambda(\mathcal{R}_n(\lambda), \lambda^2 + Y\lambda + 2Y - 3). \quad (17)$$

Its roots are

$$y(\lambda) = \frac{3 - \lambda^2}{\lambda + 2},$$

and its primitive quotient is obtained by replacing $\mathcal{R}_n$ with ψ_{p^j} .

Proof. The standard factorization of $(D_n(A) + D_n(B))/(A + B)$ over $\mathbb{Q}(\zeta_n + \zeta_n^{-1})$ gives, after substituting $B = A + 1$ and $Y = A^2 + A$, the displayed linear factors; equivalently, direct substitution shows that every $y(\lambda_k)$ is a root, and the degrees agree. The leading coefficient is $\prod_k (\lambda_k + 2) = \prod_{k=1}^{n-1} (1 + \zeta_n^k) = 1$. Grouping the trace values by exact order gives the primitive statement. ■

Define the collision factor

$$\nu(\lambda, \mu) = (\lambda + 2)(\mu + 2) - 1.$$

The elementary identity

$$y(\lambda) - y(\mu) = \frac{(\mu - \lambda)\nu(\lambda, \mu)}{(\lambda + 2)(\mu + 2)} \quad (18)$$

separates the ordinary cyclotomic discriminant from the new source of two-adic divisibility.

Theorem 4.5 (Two collision norms). *Let Λ_j be the primitive p^j trace values and let $\Lambda_{<j}$ be the nontrivial trace values of order dividing p^{j-1} . Then:*

1. *the parity of $\text{disc}(\tilde{\Pi}_{p^j})$ is the parity of the internal product*

$$N_j^{\text{int}} = \prod_{\{\lambda, \mu\} \subset \Lambda_j} \nu(\lambda, \mu);$$

2. *the parity of $\text{Res}(\tilde{\Pi}_{p^j}, \tilde{C}_{p^{j-1}})$ is the parity of the mixed product*

$$N_j^{\text{mix}} = \prod_{\lambda \in \Lambda_j, \mu \in \Lambda_{<j}} \nu(\lambda, \mu).$$

Consequently stabilization at level p^j is equivalent to the simultaneous oddness of the internal and mixed collision norms.

Proof. Equation (18) expresses the discriminant and resultant as products of ordinary trace differences times the stated collision factors. Since 2 is unramified in every odd-conductor cyclotomic field, the discriminants of $\mathcal{R}_n$ and ψ_{p^j} are odd. The factors $\lambda + 2$ are cyclotomic units of odd norm. Thus all two-adic divisibility is carried by the two displayed products. Theorem 4.3 finishes the proof. ■

Remark 4.6 (Circular-unit form). *If $\epsilon_a = \zeta^{-a/2}(1 + \zeta^a)$, then $\epsilon_a^2 = \lambda_a + 2$, so*

$$\nu(\lambda_a, \lambda_b) = \epsilon_a^2 \epsilon_b^2 - 1.$$

At a prime above two this vanishes exactly when $\epsilon_a \epsilon_b \equiv 1$. With $s_a = \zeta^{-a/2}(1 - \zeta^a)$ one has $\epsilon_a = \sigma_2(s_a)/s_a$, and the same congruence is equivalent to $s_a s_b \equiv 1$. Thus the two norms here are precisely the primitive-primitive and primitive-lower sine-unit obstructions of Section 9.

4.2 The collision involution and relative norms

Set

$$\mathbf{m}(x) = \frac{1}{x+2} - 2 = -\frac{2x+3}{x+2}.$$

It is an involution, conjugate under $U = x+2$ to $U \mapsto U^{-1}$, and

$$\nu(\lambda, \mu) = (\lambda+2)(\mu - \mathbf{m}(\lambda)).$$

Modulo two, $\mathbf{m}(x) = x^{-1}$. Thus the internal norm is odd exactly when the primitive trace polynomial is coprime modulo two to its reciprocal, and the mixed norm is odd exactly when no primitive trace is the inverse of a lower trace.

For a monic polynomial h , write

$$h^{\mathbf{m}}(X) = (X+2)^{\deg h} h(\mathbf{m}(X)).$$

The diagonal contribution to $\text{Res}(\psi_n, \psi_n^{\mathbf{m}})$ is odd for odd $n \neq 3$, because $\psi_n(-2) = \pm 1$ and $\psi_n(-1), \psi_n(-3)$ are odd. Hence

$$N_j^{\text{mix}} \text{ is odd} \iff \text{Res}\left(\psi_{p^j}, \left(\prod_{i < j} \psi_{p^i}\right)^{\mathbf{m}}\right) \text{ is odd}, \quad (19)$$

and the two-adic valuation of $\text{Res}(\psi_{p^j}, \psi_{p^j}^{\mathbf{m}})$ is twice that of the internal product.

There is also an exact one-step norm identity. If $K_i^+ = \mathbb{Q}(\zeta_{p^i} + \zeta_{p^i}^{-1})$, $\lambda = \theta + \theta^{-1}$ has exact level p^j , $\lambda_p = \theta^p + \theta^{-p}$, and $\mu \in K_{j-1}^+$, then

$$N_{K_j^+/K_{j-1}^+}((\lambda+2)(\mu+2)-1) = (\mu+2)^p (\lambda_p - D_p(\mathbf{m}(\mu))). \quad (20)$$

Multiplying this identity over lower trace orbits gives a recurrence of ordinary cyclotomic resultants. This makes the obstruction computable without constructing C_{p^j} , but it does not force its parity: on the plateau the right side can vanish modulo a chosen prime above two.

Theorem 4.7 (Mixed oddness off the plateau). *If $j > c_p$, then N_j^{mix} is odd.*

Proof. A mixed collision gives θ of exact order p^j and η of order dividing p^{j-1} such that

$$(\theta + \theta^{-1})(\eta + \eta^{-1}) = 1.$$

The inverse lower trace lies in a field of degree dividing $f_{j-1} = \text{ord}_{p^{j-1}}(2)$, and θ satisfies a quadratic over that field. Hence $f_j \mid 2f_{j-1}$. For every $j > c_p$ one has $f_j = pf_{j-1}$, which would force the odd prime p to divide 2. ■

The same degree argument says nothing about the internal norm, since both traces already lie at exact level p^j . The global degree theorem, Theorem 3.5, rules out new internal collisions off the plateau by using the five-term sparsity; collision-norm algebra alone does not.

Theorem 4.8 (Plateau inversion criterion). *Assume $j \leq c_p$, and in the common residue field put*

$$\mathcal{T}_i = \{x + x^{-1} : x \in \mu_{p^i}\}.$$

Then:

1. N_j^{mix} is even if and only if

$$\mathcal{T}_{j-1}^{-1} \cap (\mathcal{T}_j \setminus \mathcal{T}_{j-1}) \neq \emptyset;$$

2. N_j^{int} is even if and only if $\mathcal{T}_j \setminus \mathcal{T}_{j-1}$ meets its own inverse.

Proof. On the plateau all the groups μ_{p^i} lie in one finite field, and $\mathbf{m}$ reduces to inversion. Distinct pairs $\{x, x^{-1}\}$ give distinct traces, so the primitive trace values are exactly $\mathcal{T}_j \setminus \mathcal{T}_{j-1}$. The two assertions are therefore direct translations of the mixed and internal collision conditions. ■

Remark 4.9 (What was learned). *The integral route is an exact reformulation, not a parity theorem. It corrects a tempting but insufficient reduction to a single discriminant: the mixed resultant is an independent obstruction and must also be odd. Off the plateau its parity follows from degree growth, while on the plateau the problem becomes the concrete question whether a finite set of primitive trace values meets its own inverse or the inverse of the lower trace set. No general cyclotomic resultant formula determines those intersections.*

5 Arithmetic dynamics of the collision locus

Let $f = D_p$ over $\mathbb{F}_2$. Dickson composition gives $D_{p^j} = f^{\circ j}$, so the roots of D_{p^j} form the depth- j preimage tree of the fixed point 0.

Proposition 5.1 (Critically fixed structure). *The map f fixes 0. Its finite critical points are exactly the nonzero roots of D_p , each of ramification index two, and every one of them maps to 0.*

Proof. For odd p , $D_p(X) = XH_p(X)^2$, so $f'(X) = H_p(X)^2 = F_p(X)$. The square-free roots of H_p are therefore the critical points, and D_p sends all of them to zero. The point 0 is a simple fixed point because $H_p(0) = 1$. ■

Let

$$\mathcal{Z}_j = (f^{\circ j})^{-1}(0), \quad \text{Coll}_j = \mathcal{Z}_j \cap (\mathcal{Z}_j + 1).$$

Lemma 2.5 identifies Coll_j , modulo $X \mapsto X + 1$, with the repeated roots of C_{p^j} .

Proposition 5.2 (Chebyshev linearization). *The degree-two map*

$$\phi : \mathbb{G}_m \longrightarrow \mathbb{A}^1, \quad \phi(u) = u + u^{-1},$$

satisfies

$$f^{\circ j} \circ \phi = \phi \circ [u \mapsto u^{p^j}],$$

and maps μ_{p^j} two-to-one onto $\mathcal{Z}_j$, except above $\phi(1) = 0$.

Proof. This is the Dickson identity $D_{p^j}(u + u^{-1}) = u^{p^j} + u^{-p^j}$ [LMT93]. ■

Fix a generator ζ of μ_{p^j} and write $\lambda_a = \zeta^a + \zeta^{-a}$. A *collision pair* is an unordered pair $\{a, b\}$ with $\lambda_a + \lambda_b = 1$, modulo independent sign changes. It is new at level j when at least one of a, b is prime to p .

Theorem 5.3 (Covering-degree bound). *If $\{a, b\}$ is a mixed collision pair with $p \nmid a$ and $p \mid b$, then*

$$\text{ord}_{p^j}(2) \mid 2 \text{ord}_{p^{j-1}}(2).$$

Hence no mixed collision exists for $j > c_p$.

Proof. The lower trace λ_b lies in $\mathbb{F}_{2^{f_{j-1}}}$, and $\lambda_a = 1 + \lambda_b$ lies there as well. The primitive root ζ^a satisfies $T^2 + \lambda_a T + 1 = 0$, a quadratic over that field. Thus $f_j = [\mathbb{F}_2(\zeta^a) : \mathbb{F}_2] \mid 2f_{j-1}$. For $j > c_p$, $f_j = pf_{j-1}$. ■

Theorem 5.4 (Doubling-orbit factorization). *The collision pairs decompose into orbits under*

$$\{a, b\} \mapsto \{2a, 2b\}.$$

An orbit of size d contributes one irreducible factor of degree d to B_{p^j} , and every irreducible factor arises in this way.

Proof. Frobenius gives $\lambda_a^2 = \lambda_{2a}$ and therefore

$$(\lambda_a^2 + \lambda_a)^2 = \lambda_{2a}^2 + \lambda_{2a}.$$

The degree of the minimal polynomial of a repeated root is the size of its Frobenius orbit. The map from a collision pair to $Y = \lambda_a^2 + \lambda_a$ is injective modulo independent signs and interchange, because $T^2 + T + Y$ recovers the unordered pair $\{\lambda_a, \lambda_b\}$. ■

Remark 5.5 (Why dynamics also stops). *On the plateau $f_j = f_{j-1}$, so the covering-degree bound is vacuous. The monodromy is the same cyclic-by-inversion group in every residue characteristic, while Section 13 gives genuine plateau failures outside characteristic two. Dynamics therefore identifies the exact finite object to control—doubling orbits of collision pairs in one finite field—but monodromy alone cannot prove that this object is empty.*

6 The symplectic and group-theoretic attack

A separable reciprocal quartic is naturally a characteristic polynomial in rank two. This places the problem inside a well-understood finite group, but the classification controls the ambient tori rather than the required trace-one incidence.

Theorem 6.1 (Sp_4 reformulation). *A counterexample at level p^j is equivalent to a regular semisimple element $g \in \mathrm{Sp}_4(\mathbb{F}_{2^d})$, for some d , such that*

$$\mathrm{charpol}(g) = Q_Y(T) = T^4 + T^3 + YT^2 + T + 1,$$

all eigenvalues lie in μ_{p^j} , and $\mathrm{ord}(g) = p^j$. Equivalently, it is a regular semisimple element of pure p -power order and trace one.

Proof. A semisimple symplectic element has eigenvalues $\lambda, \lambda^{-1}, \mu, \mu^{-1}$, and its characteristic polynomial is

$$T^4 + (\alpha + \beta)T^3 + \alpha\beta T^2 + (\alpha + \beta)T + 1,$$

where $\alpha = \lambda + \lambda^{-1}$ and $\beta = \mu + \mu^{-1}$. In characteristic two its trace is one exactly when $\alpha + \beta = 1$, which is the five-term equation. For $p > 5$ the four eigenvalues are distinct by Lemma 1.2, and at least one has exact order p^j . Conversely, the eigenvalues of such a g give the required exponents. The realization and rational conjugacy statement for a separable reciprocal quartic are standard for symplectic groups [Wal63, Eno72]. ■

Here Y is a fixed element of $\overline{\mathbb{F}}_2$, not an indeterminate. Let $k = \mathbb{F}_2(Y) \subseteq \overline{\mathbb{F}}_2$ be the finite field it generates, put $q = |k|$, and let α be a root of $T^2 + T + Y$.

Theorem 6.2 (Maximal-torus dichotomy). *The regular class in Theorem 6.1 lies in one of the following tori.*

1. *If $\mathrm{Tr}_{k/\mathbb{F}_2}(Y) = 0$, then $\alpha \in k$ and the class lies in a product of two like SL_2 tori, of order $(q-1)^2$ or $(q+1)^2$.*

2. If $\text{Tr}_{k/\mathbb{F}_2}(Y) = 1$, then $\alpha \notin k$, the two symplectic planes are Galois conjugate, and the class lies in a cyclic torus of order $q^2 - 1$ or $q^2 + 1$.

The mixed product type $(q - 1)(q + 1)$ cannot contain the required p^j -eigenvalue configuration.

Proof. The Artin–Schreier criterion decides whether $T^2 + T + Y$ splits over k . In the split case the two reciprocal quadratics are separately defined over k . Each is either split, with eigenvalues in $k^\times$, or anisotropic, with eigenvalues in the norm-one circle of order $q + 1$. An odd prime power cannot divide both $q - 1$ and $q + 1$, so the two factors have the same type.

In the nonsplit case Frobenius over k interchanges α and $\alpha + 1$, hence interchanges the two reciprocal pairs. The four eigenvalues form one orbit in a cyclic torus of order $q^2 - 1$ or $q^2 + 1$. This is the standard rank-two torus classification [Car85, Eno72]. $\blacksquare$

On a Wieferich plateau, p^j divides the order of one of these tori, so elements of order p^j are guaranteed. What is not guaranteed is trace one. In product type that condition is

$$t_1 + t_1^{-1} + t_2 + t_2^{-1} = 1, \quad t_1, t_2 \in \mu_{p^j},$$

which is the original equation; in cyclic type the same equation holds with t_2 a Frobenius conjugate of t_1 . Thus “order is cheap, trace one is not.”

6.1 Why character theory and PSL_2 do not close the gap

For a regular element in a maximal torus T , the exact class count is a sum over Weyl orbits in $T[p^j]$ satisfying trace one. Deligne–Lusztig characters and Enomoto’s character table reorganize this sum, but the inner condition is still the same incidence on $T[p^j]$. Averaging the trace-one indicator by additive characters gives the Kloosterman expression of Section 14; its square-root error is much larger than the expected main term on a plateau.

There is also a superficially similar trace identity in $\text{PSL}_2(q) = \text{SL}_2(q)$ for q even:

$$\text{tr}(A) \text{tr}(C) = \text{tr}(AC) + \text{tr}(A^{-1}C).$$

Requiring AC of order three and $A^{-1}C$ of order two gives $\text{tr}(A) \text{tr}(C) = 1$, whereas the five-term locus is $\text{tr}(A) + \text{tr}(C) = 1$. The two loci meet only where $x^2 + x + 1 = 0$, namely in the order-three configuration excluded by $p > 5$. Triangle-group and subgroup-classification results in PSL_2 therefore address a transverse problem.

Remark 6.3 (What was learned). *The group-theoretic reformulation accurately classifies every possible ambient torus and rational conjugacy type. It also shows why merely finding p^j -torsion in Sp_4 is irrelevant. The missing statement is still that a codimension-one trace condition avoids a particular p -Sylow subset; neither maximal-torus classification nor general character estimates encode the characteristic-two additive rigidity needed for that avoidance.*

7 Characteristic zero: a complete classification

Write $\omega = e^{2\pi i/3}$ and $\zeta_n = e^{2\pi i/n}$.

Theorem 7.1. *Let u, v be complex roots of unity with*

$$1 + u + u^{-1} + v + v^{-1} = 0. \tag{21}$$

Then, up to interchanging u and v and replacing either by its inverse, exactly one of the following holds.

1. $u = \zeta_5$ and $v = \zeta_5^2$; here $\{1, u, u^{-1}, v, v^{-1}\} = \mu_5$ and $Q_Y = \Phi_5$ with $Y = 1$.
2. $u = i$ and $v = \omega$; here $Q_Y = \Phi_3\Phi_4$ with $Y = 2$.
3. $u = -1$ and $v = -\omega$; here $Q_Y = \Phi_2^2\Phi_6$ with $Y = 0$.

In particular the order pairs $(\text{ord}(u), \text{ord}(v))$ that occur are exactly $(5, 5)$, $(4, 3)$ and $(2, 6)$.

Proof. The left side of (21) is a vanishing sum of five roots of unity counted with multiplicity. Decompose it into minimal vanishing subsums. By the classification of minimal vanishing sums of weight at most nine [CJ76, Theorem 6], restated with the weight bookkeeping of [PR98, Table 1], the possible weights of a minimal vanishing sum of roots of unity are 2, 3, 5, 6, 7, $\dots$; there is none of weight 4, the unique similarity class of weight 2 is $\mathcal{R}_2 = 1 + (-1)$, the unique class of weight 3 is $\mathcal{R}_3 = 1 + \omega + \omega^2$, and the unique class of weight 5 is $\mathcal{R}_5 = 1 + \zeta_5 + \zeta_5^2 + \zeta_5^3 + \zeta_5^4$. Therefore $5 = 5$ or $5 = 2 + 3$.

Two degenerate configurations are excluded at once. If $u = 1$ then (21) reads $3 + v + v^{-1} = 0$, impossible since $|v + v^{-1}| \leq 2$. If $v = u^{\pm 1}$ then (21) reads $1 + 2(u + u^{-1}) = 0$, so $2 \cos \vartheta = -\frac{1}{2}$ for a rational multiple ϑ of 2π ; Niven's theorem [Niv56, Cor. 3.12] says the only rational values of $2 \cos \vartheta$ are $0, \pm 1, \pm 2$.

Case $5 = 5$. The multiset $\{1, u, u^{-1}, v, v^{-1}\}$ is $\eta \cdot \mu_5$ for some root of unity η . It contains 1, so $\eta^{-1} \in \mu_5$ and the multiset is μ_5 itself. Thus $\{u, u^{-1}, v, v^{-1}\} = \{\zeta_5, \zeta_5^2, \zeta_5^3, \zeta_5^4\}$, which is alternative (1).

Case $5 = 2 + 3$. Some two-element sub-multiset sums to zero, i.e. consists of x and $-x$, and the complementary three-element sub-multiset is a rotation of $\mathcal{R}_3$. We enumerate the possible pairs.

- $\{u, u^{-1}\}$: then $u^{-1} = -u$, so $u^2 = -1$ and $u = \pm i$. The remaining sum $1 + v + v^{-1} = 0$ forces $\{1, v, v^{-1}\} = \{1, \omega, \omega^2\}$, i.e. $v = \omega^{\pm 1}$. This is alternative (2). The symmetric choice $\{v, v^{-1}\}$ gives the same alternative with u and v interchanged.
- $\{1, u\}$: then $u = -1$, and the remaining multiset $\{-1, v, v^{-1}\}$ is a rotation $\eta\{1, \omega, \omega^2\}$. Containing -1 forces $\eta \in -\mu_3$, and taking $\eta = -1$ gives $\{v, v^{-1}\} = \{-\omega, -\omega^2\}$, which is consistent because $(-\omega)^{-1} = -\omega^2$. This is alternative (3); the choice $\{1, v\}$ is symmetric.
- $\{u, v\}$, $\{u, v^{-1}\}$, $\{u^{-1}, v\}$ or $\{u^{-1}, v^{-1}\}$: then $v = \pm u^{\pm 1}$ up to sign, and the complementary triple is $\{1, w, -w\}$ for some root of unity w , whose sum is $1 \neq 0$.

The identification of Q_Y in each case is a direct expansion: $(T^2 - \alpha T + 1)(T^2 - \alpha' T + 1)$ with $\alpha = -(u + u^{-1})$, $\alpha' = -(v + v^{-1})$, $\alpha + \alpha' = 1$ and $Y = 2 + \alpha\alpha'$. ■

Theorem 7.1 was verified by exhaustive search over all pairs of roots of unity of order dividing $420 = 2^2 \cdot 3 \cdot 5 \cdot 7$; the only order pairs found were $(5, 5)$, $(3, 4)$ and $(2, 6)$.

Corollary 7.2. *Let $p > 5$ be prime, $j \geq 1$, $\zeta = \zeta_{p^j}$. Then*

$$\Lambda_{a,b} := 1 + \zeta^a + \zeta^{-a} + \zeta^b + \zeta^{-b} \neq 0 \quad \text{for all } a, b \in \mathbb{Z}.$$

Proof. By Theorem 7.1 any vanishing instance has $\text{ord}(\zeta^a) \in \{2, 3, 4, 5, 6\}$, which is impossible for a divisor of p^j with $p > 5$ unless $\zeta^a = 1$; but the pattern $(u, v) = (1, *)$ does not occur in the classification. Alternatively, and more directly, Lam and Leung's main theorem [LL00, Main Theorem] says that the weights of vanishing sums of p^j -th roots of unity form the numerical semigroup $\mathbb{N}p$; since $0 < 5 < p$, the weight 5 is not attained. ■

Remark 7.3. *Corollary 7.2 is the reason Problem 1.1 is not a statement about exact vanishing sums at all. The characteristic-zero equation has no solutions whatsoever for $p > 5$; the entire content of the problem is that $\Lambda_{a,b}$, a nonzero algebraic integer, might nevertheless be divisible by a prime above 2.*

8 Cyclotomic units: an unconditional family

Corollary 7.2 says $\Lambda_{a,b} \neq 0$. If in addition $\Lambda_{a,b}$ is a *unit*, then it is nonzero in every residue field and (4) is impossible in every characteristic. This section determines exactly when that happens for structural reasons.

For $r \geq 1$ put

$$G_r(x) = x^{2r} + x^{r+1} + x^r + x^{r-1} + 1 \in \mathbb{Z}[x], \quad \text{so} \quad G_r(x) = x^r (1 + x + x^{-1} + x^r + x^{-r}).$$

Theorem 8.1. *G_r is a product of cyclotomic polynomials if and only if $r \in \{2, 3\}$, in which case*

$$G_2 = \Phi_5, \quad G_3 = \Phi_5 \Phi_6.$$

Proof. The two displayed factorizations are direct expansions. Conversely suppose every root of G_r is a root of unity. If u is a root, then $u \neq 0$ and $1 + u + u^{-1} + u^r + u^{-r} = 0$, so (u, u^r) is one of the pairs of Theorem 7.1. Alternative (2) is impossible because $\text{ord}(u) = 4$ and $\text{ord}(u^r) = 3$ cannot hold simultaneously with u^r a power of u , and likewise with the roles reversed. Alternative (1) forces $\text{ord}(u) = 5$, alternative (3) forces $\text{ord}(u) \in \{2, 6\}$; and $u = -1$ is excluded because

$$G_r(-1) = 1 + (-1)^{r+1} + (-1)^r + (-1)^{r-1} + 1 = 2 + (-1)^{r-1} \in \{1, 3\}.$$

Hence every root of G_r has order 5 or 6, so $G_r = \Phi_5^{e_5} \Phi_6^{e_6}$ and $2r = 4e_5 + 2e_6$.

We show $e_5 \leq 1$ and $e_6 \leq 1$. Set $\Lambda(x) = 1 + x + x^{-1} + x^r + x^{-r}$, so that $G_r = x^r \Lambda$ and at a root of G_r the derivative G'_r vanishes if and only if Λ' does. Now

$$\Lambda'(x) = 1 - x^{-2} + r(x^{r-1} - x^{-r-1}).$$

If $\text{ord}(u) = 5$ then, by alternative (1), $u^r = u^{\pm 2}$; replacing u by u^{-1} if necessary, $u^r = u^2$, so $u^{r-1} = u$ and $u^{-r-1} = u^{-3} = u^2$, giving $\Lambda'(u) = 1 - u^3 + r(u - u^2)$. As $1, u, u^2, u^3$ is a $\mathbb{Q}$ -basis of $\mathbb{Q}(\zeta_5)$, this is nonzero. If $\text{ord}(u) = 6$ then $u^r = -1$ by alternative (3), so $u^{r-1} = -u^{-1} = u^{-r-1}$ and the bracket vanishes, leaving $\Lambda'(u) = 1 - u^{-2} \neq 0$ because $u^2 \neq 1$. Hence G_r has no repeated roots, $e_5, e_6 \leq 1$, and $2r \leq 4 + 2 = 6$. Finally $r = 1$ is excluded because $G_1 = 2x^2 + x + 2$ is not monic. ■

Corollary 8.2. *Let $p > 5$ be prime and $j \geq 1$, and let a, b be integers with*

$$b \equiv \pm 2a, \pm 3a \quad \text{or} \quad a \equiv \pm 2b, \pm 3b \pmod{p^j}.$$

Then $\Lambda_{a,b}$ is a unit of $\mathbb{Z}[\zeta_{p^j}]$ if $a \not\equiv 0$, and $\Lambda_{a,b} = 5$ if $a \equiv b \equiv 0$. In the unit case, (4) has no solution with these exponents in any residue characteristic. In the degenerate case $\Lambda_{a,b} = 5$, it can vanish only in characteristic 5. In particular, neither case gives a solution in characteristic 2.

Proof. Since $\Lambda_{a,b} = \Lambda_{b,a} = \Lambda_{a,-b} = \Lambda_{-a,b}$, it suffices to treat $b = 2a$ and $b = 3a$. Put $\xi = \zeta^a$, of exact order p^i say. If $i = 0$ then $\Lambda = 5$. If $i \geq 1$ then by Theorem 8.1

$$\Lambda_{a,2a} = \xi^{-2} \Phi_5(\xi), \quad \Lambda_{a,3a} = \xi^{-3} \Phi_5(\xi) \Phi_6(\xi).$$

By Apostol's evaluation of resultants of cyclotomic polynomials [Apo70], $\text{Res}(\Phi_m, \Phi_n) = q^{\varphi(n)}$ when m/n is a power of a prime q and $\text{Res}(\Phi_m, \Phi_n) = 1$ otherwise. With $m = p^i \geq 7$ and $n \in \{5, 6\}$ the quotient m/n is not an integer, so both resultants equal 1. Hence $N_{\mathbb{Q}(\xi)/\mathbb{Q}}(\Phi_5(\xi)) = \pm 1$ and $N_{\mathbb{Q}(\xi)/\mathbb{Q}}(\Phi_6(\xi)) = \pm 1$, and $\Lambda_{a,b}$ is a unit. A unit has odd norm, so its image in $\mathbb{Z}[\zeta_{p^j}]/\mathfrak{p}$ is nonzero for every prime $\mathfrak{p}$. ■

Remark 8.3. *The two identities behind Corollary 8.2 are the cyclotomic unit identities*

$$1 + \xi + \xi^{-1} + \xi^2 + \xi^{-2} = \xi^{-2} \frac{\xi^5 - 1}{\xi - 1}, \quad 1 + \xi + \xi^{-1} + \xi^3 + \xi^{-3} = \xi^{-3} \frac{\xi^5 - 1}{\xi - 1} \cdot \frac{\xi^6 - 1}{\xi^3 - 1} \cdot \frac{\xi - 1}{\xi^2 - 1}.$$

Theorem 8.1 says that these are the only two members of the family, so the cyclotomic unit method removes at most 8 of the $\varphi(p^j)$ possible ratios b/a — exactly 8 when the classes $\{\pm 2^{\pm 1}\}$ and $\{\pm 3^{\pm 1}\}$ are distinct modulo p^j , and only 4 when they coincide, as they do modulo 7 — and cannot remove more.

Example 8.4. For $p^j = 49$ the ratio classes $\{\pm r^{\pm 1}\}$ that consist of units are $\{2, 24, 25, 47\}$ and $\{3, 16, 33, 46\}$, exactly as predicted; see Table 3. The class $\{13, 15, 34, 36\}$ also has relative norm 1, but sporadically: $G_{13} = \Phi_5 \cdot H$ with H of degree 22 and Mahler measure $\approx 0.2507 > 0$, and $\text{Res}(\Phi_n, G_{13}) \neq \pm 1$ for $n = 13, 17, 23, 29, 31$.

9 Reduction modulo a prime above two

Fix $p > 5$, $j \geq 1$, and set

$$K = \mathbb{Q}(\zeta_{p^j}), \quad K^+ = \mathbb{Q}(\zeta_{p^j} + \zeta_{p^j}^{-1}), \quad f = \text{ord}_{p^j}(2), \quad g = \frac{\varphi(p^j)}{f}.$$

Since p is odd, 2 is unramified in K and splits into g distinct primes, each with residue field $\mathbb{F}_{2^f}$ and decomposition group $\langle \sigma_2 \rangle$, where $\sigma_t(\zeta) = \zeta^t$ [Was97, Ch. 2]. Reduction modulo any prime $\mathfrak{p} \mid 2$ restricts to a group isomorphism $\mu_{p^j} \rightarrow \mu_{p^j}(\mathbb{F}_{2^f})$.

Proposition 9.1. *Let $\Sigma_{a,b} = \{\mathfrak{p} \mid 2 : \mathfrak{p} \mid \Lambda_{a,b}\}$. Then:*

1. (4) holds for some θ of exact order p^j and the given (a, b) if and only if $\Sigma_{a,b} \neq \emptyset$;
2. $\Lambda_{a,b} \in \mathbb{Z}[\zeta_{p^j} + \zeta_{p^j}^{-1}]$, so $N_{K/\mathbb{Q}}(\Lambda_{a,b}) = N_{K^+/\mathbb{Q}}(\Lambda_{a,b})^2$;
3. $v_2(N_{K/\mathbb{Q}}(\Lambda_{a,b})) \geq f \cdot |\Sigma_{a,b}|$, with equality when every $v_{\mathfrak{p}}(\Lambda_{a,b}) \leq 1$;
4. writing $r = ba^{-1}$ when $\gcd(a, p) = 1$,

$$N_{K/\mathbb{Q}}(\Lambda_{a,b}) = \pm \text{Res}(\Phi_{p^j}(x), G_r(x)).$$

In particular, Problem 1.1 has an affirmative answer for (p, j) if and only if $\text{Res}(\Phi_{p^j}, G_r)$ is odd for every nonzero $r \in \mathbb{Z}/p^j\mathbb{Z}$. (A counterexample has $p \nmid a$ or $p \nmid b$; after interchanging a and b we may assume $\gcd(a, p) = 1$, and then $r = ba^{-1}$ is a well-defined nonzero element of $\mathbb{Z}/p^j\mathbb{Z}$, not necessarily a unit.)

Proof. (1) is the reduction isomorphism together with the fact that the g primes above 2 are permuted transitively by $\text{Gal}(K/\mathbb{Q})$, and $\sigma_t(\Lambda_{a,b}) = \Lambda_{ta, tb}$. (2) holds because the five summands are stable under $\zeta \mapsto \zeta^{-1}$, and $[K : K^+] = 2$. (3) is the norm formula $v_2(N_{K/\mathbb{Q}}(x)) = \sum_{\mathfrak{p} \mid 2} f v_{\mathfrak{p}}(x)$ valid because 2 is unramified with residue degree f at every $\mathfrak{p}$. (4) follows from $\Lambda_{a,b} = \zeta^{-ra} G_r(\zeta^a)$ and $N_{K/\mathbb{Q}}(\zeta) = \pm 1$, after the substitution $\zeta \mapsto \zeta^a$, which permutes the primitive p^j -th roots of unity; the fact that only the ratio r matters is Lemma 1.3. ■

9.1 Circular units and the exact local obstruction

The parity question can be made slightly sharper by replacing the five-term sum by products of circular units. Write $K_j = \mathbb{Q}(\zeta_j)$, $K_j^+ = \mathbb{Q}(\zeta_j + \zeta_j^{-1})$ with $\zeta_j = \zeta_{p^j}$, put $n = p^j$, and choose the integer $h_j = (n+1)/2$, so that $2h_j \equiv 1 \pmod{n}$. For $a \in \mathbb{Z}/n\mathbb{Z}$ define

$$\epsilon_{j,a} = \zeta_j^{-h_j a} (1 + \zeta_j^a).$$

Also put

$$s_{j,a} = \zeta_j^{-h_j a} (1 - \zeta_j^a).$$

Thus $\epsilon_{j,-a} = \epsilon_{j,a}$ and

$$\epsilon_{j,a} = \zeta_j^{a/2} + \zeta_j^{-a/2}, \quad \epsilon_{j,a}^2 = \lambda_a + 2, \quad \lambda_a = \zeta_j^a + \zeta_j^{-a}.$$

If $a \not\equiv 0 \pmod{n}$ then $\epsilon_{j,a}$ is a circular unit: indeed $1 + \xi = (1 - \xi^2)/(1 - \xi)$ for $\xi = \zeta_j^a$ of odd order, and $\Phi_{\text{ord}(\xi)}(-1) = 1$. For such a , the auxiliary element $s_{j,a}$ is a p -unit, integral and a unit at every prime above 2, and for $\sigma_2(\zeta_j) = \zeta_j^2$ one has the Hilbert–90 identity

$$\frac{\sigma_2(s_{j,a})}{s_{j,a}} = \zeta_j^{-h_j a} (1 + \zeta_j^a) = \epsilon_{j,a}. \quad (22)$$

Thus each $\epsilon_{j,a}$ has local norm one for the unramified decomposition extension at 2, and is not merely a circular unit but a specified local coboundary.

Lemma 9.2 (norm compatibility). *For $j > 1$,*

$$N_{K_j/K_{j-1}}(s_{j,a}) = s_{j-1,\bar{a}}, \quad N_{K_j/K_{j-1}}(\epsilon_{j,a}) = \epsilon_{j-1,\bar{a}} \quad (p \nmid a),$$

where $\bar{a}$ denotes reduction modulo p^{j-1} , and

$$s_{j,pa} = s_{j-1,a}, \quad \epsilon_{j,pa} = \epsilon_{j-1,a}.$$

Proof. The conjugates over K_{j-1} send ζ_j to $\zeta_j^{1+tp^{j-1}}$, $0 \leq t < p$. Hence

$$N_{K_j/K_{j-1}}(1 + \zeta_j^a) = 1 + \zeta_{j-1}^a$$

for $p \nmid a$, by evaluating $T^p - \zeta_{j-1}^a$ at $T = -1$; the same evaluation at $T = 1$ gives $N_{K_j/K_{j-1}}(1 - \zeta_j^a) = 1 - \zeta_{j-1}^a$. Also $N_{K_j/K_{j-1}}(\zeta_j) = \zeta_{j-1}$, and $h_j \equiv h_{j-1} \pmod{p^{j-1}}$, giving the first identity. The second follows at once from $\zeta_j^p = \zeta_{j-1}$ and the same congruence for h_j . ■

Proposition 9.3 (Hilbert–90 reformulation). *Let $\mathfrak{P} \mid 2$ be a prime of K_j and let $a, b \in \mathbb{Z}/n\mathbb{Z}$ be nonzero. Put $w_{j,a,b} = s_{j,a}s_{j,b}$. Then the following are equivalent:*

$$\epsilon_{j,a}\epsilon_{j,b} \equiv 1 \pmod{\mathfrak{P}}, \quad w_{j,a,b} \equiv 1 \pmod{\mathfrak{P}},$$

and

$$1 + \zeta_j^{(a+b)/2} + \zeta_j^{-(a+b)/2} + \zeta_j^{(a-b)/2} + \zeta_j^{-(a-b)/2} \equiv 0 \pmod{\mathfrak{P}}.$$

Consequently a conductor- p^j counterexample to Problem 1.1 is exactly a congruence $\epsilon_{j,a}\epsilon_{j,b} \equiv 1$ modulo one prime above 2, with at least one of a, b prime to p .

Proof. Since σ_2 lies in the decomposition group at every prime above 2, reduction modulo $\mathfrak{P}$ sends $\sigma_2(x)$ to $\bar{x}^2$. By (22),

$$\overline{\epsilon_{j,a}\epsilon_{j,b}} = \frac{\overline{w_{j,a,b}}^2}{\overline{w_{j,a,b}}} = \overline{w_{j,a,b}},$$

because $w_{j,a,b}$ is a unit at $\mathfrak{P}$. Hence $\epsilon_{j,a}\epsilon_{j,b} \equiv 1$ if and only if $w_{j,a,b} \equiv 1$. Equivalently, $\sigma_2(w_{j,a,b})/w_{j,a,b} \equiv 1$ forces the residue of $w_{j,a,b}$ to be Frobenius-fixed, hence to lie in $\mathbb{F}_2^\times = \{1\}$.

Expanding the product gives

$$\epsilon_{j,a}\epsilon_{j,b} = \lambda_{(a+b)/2} + \lambda_{(a-b)/2}.$$

In the residue field of $\mathfrak{P}$, subtracting 1 is the same as adding 1, so the displayed equivalence is immediate. The change of variables from the original exponents A, B is $a = A + B$, $b = A - B$. Since 2 is invertible modulo p^j , A and B are both divisible by p if and only if a and b are both divisible by p . ■

For the local cohomological interpretation, let $L = K_{j,\mathfrak{P}}$, let $U = \mathcal{O}_L^\times$ and $U^1 = 1 + \mathfrak{P}$, and put $D = \langle \sigma_2 \rangle$. The coboundary map $\delta(u) = \sigma_2(u)/u$ induces on the residue sequence $1 \rightarrow U^1 \rightarrow U \rightarrow \mathbb{F}_{2^f}^\times \rightarrow 1$ the map $\bar{u} \mapsto \bar{u}^2/\bar{u} = \bar{u}$. Therefore

$$\delta^{-1}(U^1) = U^1,$$

and $H^1(D, U) = H^1(D, U^1) = 0$ in this unramified local situation. Thus Hilbert 90 merely says that the norm-one unit $\epsilon_{j,a}\epsilon_{j,b}$ is a coboundary; the extra parity condition is exactly that the chosen p -unit

primitive $w_{j,a,b} = s_{j,a}s_{j,b}$ already lies in the principal local unit group U^1 . In ray-class language, if $\mathfrak{l}_j^+$ is the unique prime of K_j^+ above p and $v_p(a) = \alpha$, $v_p(b) = \beta$, then

$$(w_{j,a,b}) = (\mathfrak{l}_j^+)^{(p^\alpha + p^\beta)/2} \quad \text{in } K_j^+.$$

Thus a primitive–primitive collision is a generator of $\mathfrak{l}_j^+$ that is 1 modulo one prime above 2; a primitive–lower collision is the analogous ray-class condition for $(\mathfrak{l}_j^+)^{(1+p^\beta)/2}$ with $\beta \geq 1$, with the generator restricted to the sine-circular form $s_{j,a}s_{j,b}$.

This isolates exactly the two collision norms of Theorem 4.5. Let

$$P_j = (\mathbb{Z}/p^j\mathbb{Z})^\times / \{\pm 1\}, \quad L_j = (p\mathbb{Z}/p^j\mathbb{Z} \setminus \{0\}) / \{\pm 1\}.$$

Define

$$N_j^{\text{pp}} = \prod_{\substack{\{a,b\} \subset P_j \\ a \neq b}} N_{K_j^+/\mathbb{Q}}(\epsilon_{j,a}^2 \epsilon_{j,b}^2 - 1), \quad N_j^{\text{pl}} = \prod_{\substack{a \in P_j \\ b \in L_j}} N_{K_j^+/\mathbb{Q}}(\epsilon_{j,a}^2 \epsilon_{j,b}^2 - 1).$$

Because the residue fields above 2 have characteristic two, $x^2 - 1 \equiv 0$ for a unit x if and only if $x \equiv 1$. Therefore Problem 1.1 at level p^j is equivalent to the simultaneous oddness of N_j^{pp} and N_j^{pl} . In the notation $\nu(\lambda, \mu) = (\lambda + 2)(\mu + 2) - 1$ of the reciprocal-quartic note, these are exactly the primitive–primitive and primitive–lower products, since $\lambda_a + 2 = \epsilon_{j,a}^2$.

Now suppose $2 \leq j \leq c_p$, so we are on the base-2 plateau. Then $\text{ord}_{p^j}(2) = \text{ord}_{p^{j-1}}(2)$, and every prime above 2 in K_{j-1} splits completely in K_j . If $\mathfrak{P}$ is a prime of K_{j-1} above 2, then for $p \nmid a$,

$$\prod_{\Omega|\mathfrak{P}} \overline{s_{j,a}}^{(\Omega)} = \overline{s_{j-1,\bar{a}}}^{(\mathfrak{P})}, \quad \prod_{\Omega|\mathfrak{P}} \overline{\epsilon_{j,a}}^{(\Omega)} = \overline{\epsilon_{j-1,\bar{a}}}^{(\mathfrak{P})},$$

and, for lower-level indices,

$$\prod_{\Omega|\mathfrak{P}} \overline{s_{j,pb}}^{(\Omega)} = \left(\overline{s_{j-1,\bar{b}}}^{(\mathfrak{P})} \right)^p, \quad \prod_{\Omega|\mathfrak{P}} \overline{\epsilon_{j,pb}}^{(\Omega)} = \left(\overline{\epsilon_{j-1,\bar{b}}}^{(\mathfrak{P})} \right)^p.$$

These identities are just Lemma 9.2 reduced modulo $\mathfrak{P}$. Thus plateau splitting supplies only a product constraint over the p primes above $\mathfrak{P}$: if $w_{j,a,b} \equiv 1$ (equivalently $\epsilon_{j,a}\epsilon_{j,b} \equiv 1$) at one of them, the remaining $p-1$ conjugate factors must multiply to the prescribed lower-level p -unit. There is no contradiction from splitting alone; a proof must control the *semi-local* reduction of these particular sine and circular units.

Equivalently, let $\mathcal{E}_j^+$ be the p -unit group of K_j^+ and let $\mathcal{C}_j^+$ be Sinnott’s circular-unit group. The semi-local reduction map

$$\rho_j : \mathcal{E}_j^+ \longrightarrow \prod_{\mathfrak{q}|2} (\mathcal{O}_{K_j^+}/\mathfrak{q})^\times$$

is the relevant finite quotient. The desired parity theorem says precisely that no sine p -unit $s_{j,a}s_{j,b}$, with $a \in P_j$ and $b \in P_j \cup L_j$, lies in any one-prime principal-unit kernel $\mathcal{E}_j^+ \cap (1 + \mathfrak{q})$. Applying the coboundary $\delta = \sigma_2 - 1$ sends this finite set into $\mathcal{C}_j^+$ as the circular units $\epsilon_{j,a}\epsilon_{j,b}$; the sine lift is essential, because $\delta^{-1}(1 + \mathfrak{q}) = 1 + \mathfrak{q}$ only after using the characteristic-two residue field. The standard ray-class sequence for modulus $2\mathcal{O}_{K_j^+}$ identifies

$$\left(\prod_{\mathfrak{q}|2} (\mathcal{O}_{K_j^+}/\mathfrak{q})^\times \right) / \rho_j(\mathcal{O}_{K_j^+}^\times)$$

with the kernel of the ray class group modulo 2 over the ordinary class group. Replacing full units by circular units gives a natural surjection from the corresponding circular cokernel to this ray-class kernel, with kernel a quotient of $\mathcal{O}_{K_j^+}^\times / \mathcal{C}_j^+$ (whose size is governed by Sinnott’s formula), while the Stickelberger–Jacobi-sum formulas describe character sums and ideal classes. Those tools do not by themselves decide the required pointwise avoidance in the finite residue fields: for a chosen $\mathfrak{q}$, the congruence is equivalent to $\chi(\epsilon_{j,a})\chi(\epsilon_{j,b}) = 1$ for every multiplicative character χ of the residue field, and the characters of order prime to p are needed as well as the p -primary Jacobi-sum data.

9.2 What the reciprocal symmetry buys

Proposition 9.4. *Let $\Sigma = \Sigma_{a,b} \neq \emptyset$, and assume $\gcd(a, p) = 1$, which holds after possibly interchanging a and b in any counterexample to Problem 1.1. Let*

$$H = \{t \in (\mathbb{Z}/p^j\mathbb{Z})^\times : \{\pm ta, \pm tb\} = \{\pm a, \pm b\}\}$$

be the stabilizer of $\Lambda_{a,b}$ in $\text{Gal}(K/\mathbb{Q})$. Then Σ is a union of H -orbits, $-1 \in H$, and $H \subseteq \{t : t^4 = 1\}$, so $|H| \leq 4$. Consequently

$$|\Sigma| \geq \frac{|H|}{|H \cap \langle 2 \rangle|},$$

and in particular $|\Sigma| \geq 2$ whenever $-1 \notin \langle 2 \rangle$ in $(\mathbb{Z}/p^j\mathbb{Z})^\times$, i.e. whenever $\rho_2(p^j) = \text{ord}_{p^j}(2)$.

Proof. If $t \in H$ then $\sigma_t(\Lambda_{a,b}) = \Lambda_{a,b}$, so $\mathfrak{p} \mid \Lambda_{a,b}$ implies $\sigma_t(\mathfrak{p}) \mid \Lambda_{a,b}$; thus Σ is H -stable. Clearly $-1 \in H$. Let $t \in H$. If $ta \equiv \pm a$, then $t \equiv \pm 1$ because a is invertible. Otherwise $ta \equiv \pm b$ and $tb \equiv \pm a$; the second relation shows b is invertible, and multiplying the two gives $t^2 ab \equiv \pm ab$, hence $t^2 \equiv \pm 1$ and $t^4 \equiv 1$. Since $(\mathbb{Z}/p^j\mathbb{Z})^\times$ is cyclic, $|\{t : t^4 = 1\}| = \gcd(4, p-1) \leq 4$. The orbit of $\mathfrak{p}$ under H has size $|H| / |H \cap D_{\mathfrak{p}}|$ with $D_{\mathfrak{p}} = \langle 2 \rangle$. If $-1 \notin \langle 2 \rangle$ then $-1 \in H \setminus \langle 2 \rangle$, so $|H| / |H \cap \langle 2 \rangle| \geq 2$. ■

Example 9.5. *For $p = 17$, $j = 1$, one has $\text{ord}_{17}(2) = 8$, $\varphi(17) = 16$, $g = 2$ and $-1 = 2^4 \in \langle 2 \rangle$. The ratio class of $r = 13$ gives $\text{Res}(\Phi_{17}, G_{13}) = 256 = 2^8$, i.e. $|\Sigma| = 1$; the relation $1 + \theta^3 + \theta^{-3} + \theta^5 + \theta^{-5} = 0$ indeed holds in $\mathbb{F}_{256}$ for a suitable θ of order 17, and $|S_{17}| = 16$, matching $d(16) = 8$.*

For $p = 31$, $j = 1$, one has $\text{ord}_{31}(2) = 5$, $\varphi(31) = 30$, $g = 6$, and $-1 \notin \langle 2 \rangle = \{1, 2, 4, 8, 16\}$. The class of $r = 5$ gives $\text{Res}(\Phi_{31}, G_5) = 1024 = 2^{10} = 2^{5 \cdot 2}$, i.e. $|\Sigma| = 2$, in exact agreement with Proposition 9.4. Here $|S_{31}| = 40$ and $d(30) = 20$.

10 Sine residues, ray classes and power residues

The reformulations of Section 9 say that a counterexample is a congruence $s_{j,a}s_{j,b} \equiv 1$ at one prime above 2. This section carries that out concretely. We compute the residues of the sine p -units, prove that the residue map is injective on $\pm$ -classes, and use this to settle unconditionally every ratio lying in the decomposition group at 2 (Theorem 10.9). We then make the ray-class formulation explicit as a family of power-residue conditions, and determine the p -primary member of that family — the Stickelberger–Jacobi part — completely (Theorem 10.15). The outcome is a genuine obstruction, strong enough to prove $d(p^j - 1) = 0$ for every prime in an explicit family, conditional only on one power-residue nonvanishing that is verified in every computed case (Theorem 10.17). Section 10.6 then removes that hypothesis by an additive (Artin–Schreier) argument (Theorem 10.21, Corollary 10.23). The obstruction is vacuous when $\text{ord}_p(2)$ is even, and for odd order its descent fails on the Wieferich plateau (Proposition 10.27).

Throughout this section $p > 5$, $n = p^j$, $f = \text{ord}_n(2)$, $q = 2^f$, $\mathfrak{P} \mid 2$ is a fixed prime of $K_j = \mathbb{Q}(\zeta_n)$, and $\theta = \zeta_n \bmod \mathfrak{P}$ has exact order n in $\mathbb{F}_q^\times$. For $x \in \mathbb{Z}/n\mathbb{Z}$ we write

$$\bar{\lambda}_x = \theta^x + \theta^{-x} \in \mathbb{F}_q,$$

the reduction of $\lambda_x = \zeta_n^x + \zeta_n^{-x}$.

10.1 The residue of a sine unit is a trace

Lemma 10.1. *Let $h = (n+1)/2$, so $2h \equiv 1 \pmod{n}$. For every $a \in \mathbb{Z}/n\mathbb{Z}$ with $a \not\equiv 0$,*

$$\overline{s_{j,a}} = \overline{\epsilon_{j,a}} = \bar{\lambda}_{ha}.$$

Moreover, for all $x, y \in \mathbb{Z}/n\mathbb{Z}$,

$$\bar{\lambda}_{2x} = \bar{\lambda}_x^2, \quad \bar{\lambda}_{-x} = \bar{\lambda}_x, \quad \bar{\lambda}_x \bar{\lambda}_y = \bar{\lambda}_{x+y} + \bar{\lambda}_{x-y}. \quad (23)$$

Proof. From $2h \equiv 1$ we get $1 - h \equiv h$, hence $\overline{s_{j,a}} = \theta^{-ha}(1 + \theta^a) = \theta^{-ha} + \theta^{a-ha} = \theta^{-ha} + \theta^{ha}$. Reducing (22) modulo $\mathfrak{P}$ and using that σ_2 acts as the Frobenius on the residue field gives $\overline{\epsilon_{j,a}} = \overline{s_{j,a}}^2 / \overline{s_{j,a}} = \overline{s_{j,a}}$, because $s_{j,a}$ is a unit at $\mathfrak{P}$. The three rules in (23) are the Frobenius, the substitution $x \mapsto -x$, and multiplying out. ■

Lemma 10.2 (fibre lemma). *For $x, y \in \mathbb{Z}/n\mathbb{Z}$:*

1. $\bar{\lambda}_x = 0$ if and only if $x \equiv 0$;
2. $\bar{\lambda}_x \neq 1$ for every x ;
3. $\bar{\lambda}_x = \bar{\lambda}_y$ if and only if $y \equiv \pm x \pmod{n}$.

Consequently $\mathbb{F}_2(\bar{\lambda}_x) = \mathbb{F}_{2^d}$, where d is the order of 2 in $(\mathbb{Z}/m\mathbb{Z})^\times / \{\pm 1\}$ and $m = n / \gcd(x, n)$ is the exact order of θ^x .

Proof. In characteristic two $(T - \theta^x)(T - \theta^{-x}) = T^2 + \bar{\lambda}_x T + 1$, so $\bar{\lambda}_x$ determines the unordered pair $\{\theta^x, \theta^{-x}\}$, and (3) follows because reduction is injective on μ_n . For (1), $\bar{\lambda}_x = 0$ forces $\theta^{2x} = 1$, hence $x \equiv 0$ because n is odd. For (2), $\bar{\lambda}_x = 1$ forces $\theta^{2x} + \theta^x + 1 = 0$, so θ^x has order 3, impossible for $p > 3$. The last statement is (3) applied to $y = 2^k x$: one has $\bar{\lambda}_x^{2^k} = \bar{\lambda}_{2^k x} = \bar{\lambda}_x$ if and only if $2^k \equiv \pm 1 \pmod{m}$. ■

Proposition 10.3 (product dictionary). *Put $x = h(A + B)$ and $y = h(A - B)$, a bijection of $(\mathbb{Z}/n\mathbb{Z})^2$ onto itself with inverse $A = x + y$, $B = x - y$. Then*

$$\Lambda_{A,B} \equiv 0 \pmod{\mathfrak{P}} \iff \bar{\lambda}_x \bar{\lambda}_y = 1,$$

the excluded degenerate cases $A \equiv 0$, $B \equiv 0$, $B \equiv \pm A$ correspond to $y \equiv -x$, $y \equiv x$, $xy \equiv 0$, and $p \mid A, B$ if and only if $p \mid x, y$. The ratios correspond by the Möbius involution

$$r_{\text{sum}} = B/A \iff r_{\text{prod}} = y/x = \frac{1 - r_{\text{sum}}}{1 + r_{\text{sum}}}.$$

In particular Problem 1.1 at level p^j asserts that there is no pair $x, y \in \mathbb{Z}/n\mathbb{Z}$, not both divisible by p , with $\bar{\lambda}_x \bar{\lambda}_y = 1$.

Proof. Immediate from (23): $\bar{\lambda}_x \bar{\lambda}_y = \bar{\lambda}_{x+y} + \bar{\lambda}_{x-y} = \bar{\lambda}_A + \bar{\lambda}_B$, and $\Lambda_{A,B} = 1 + \lambda_A + \lambda_B$. The statement on ratios is the computation $y/x = (1 - B/A)/(1 + B/A)$; h is invertible, so $p \mid x, y$ iff $p \mid A, B$. ■

We record two exact identities among the residues, both of which are reductions of norm relations and both of which will be used below.

Lemma 10.4.

$$\prod_{t \in (\mathbb{Z}/n\mathbb{Z})^\times} \bar{\lambda}_t = 1, \quad \prod_{t \in H_\beta} \bar{\lambda}_{xt} = \bar{\lambda}_{p^\beta x} \quad (1 \leq \beta \leq j),$$

where $H_\beta = 1 + p^{j-\beta}\mathbb{Z}/p^j\mathbb{Z}$ is the subgroup of order p^β of $(\mathbb{Z}/n\mathbb{Z})^\times$ and $\gcd(x, p) = 1$.

Proof. Write $\bar{\lambda}_t = \theta^{-t}(1 + \theta^{2t})$. Pairing t with $n - t$ gives $\sum_t t \equiv 0 \pmod{n}$, while $\prod_t (1 + \theta^{2t}) = \prod_u (1 + \theta^u) = \Phi_n(-1) = 1$. This is the first identity. For the second, $2xt$ runs over $2x + p^{j-\beta}\mathbb{Z}/p^j\mathbb{Z}$ as t runs over H_β , so $\prod_t (1 + \theta^{2xt}) = \prod_{w \bmod p^\beta} (1 + \theta^{2x}\eta^w) = 1 + \theta^{2p^\beta x}$ with $\eta = \theta^{p^{j-\beta}}$ of order p^β ; and $\sum_{t \in H_\beta} xt \equiv p^\beta x \pmod{p^j}$. ■

10.2 The semi-local residue module and a no-go theorem

Put

$$\bar{G} = (\mathbb{Z}/n\mathbb{Z})^\times / \{\pm 1\}, \quad \bar{f} = \rho_2(n), \quad \bar{D} = \langle 2 \rangle \subseteq \bar{G}, \quad g^+ = \frac{\varphi(n)}{2\bar{f}},$$

and $E = \mathbb{F}_{2^{\bar{f}}}$. The g^+ primes of K_j^+ above two are permuted simply transitively by $\bar{G}/\bar{D}$, and their residue fields are all E . Let

$$M = \prod_{\mathfrak{q}|2} (\mathcal{O}_{K_j^+}/\mathfrak{q})^\times.$$

Theorem 10.5 (Semi-local module). *There is an isomorphism of $\bar{G}$ -modules*

$$M \cong \mathbb{Z}[\bar{G}] / (\sigma_2 - 2)\mathbb{Z}[\bar{G}].$$

Reduction at one chosen prime is evaluation on the decomposition group: it kills the cosets outside $\bar{D}$ and sends σ_2^i to multiplication by 2^i in $\mathbb{Z}/(2^{\bar{f}} - 1)\mathbb{Z}$.

Proof. As a permutation module on the primes above two,

$$M \cong \text{Ind}_{\bar{D}}^{\bar{G}} E^\times.$$

Choose a generator γ of $E^\times$. The map

$$\mathbb{Z}[\bar{D}] \longrightarrow E^\times, \quad \sum_i a_i \sigma_2^i \longmapsto \gamma^{\sum_i a_i 2^i}$$

is surjective with kernel generated by $\sigma_2 - 2$, since

$$\mathbb{Z}[X] / (X^{\bar{f}} - 1, X - 2) \cong \mathbb{Z} / (2^{\bar{f}} - 1)\mathbb{Z}.$$

Inducing to $\bar{G}$ proves the first assertion. Projection to the trivial coset gives the stated one-prime evaluation map. ■

The structural relations satisfied by the true residues are the Frobenius and sign rules in (23), the distribution relations for circular units [Sin78, Sin80, KL81], and their norm consequence. Abstract them as follows.

Definition 10.6. *An admissible system is a map*

$$m : (\mathbb{Z}/n\mathbb{Z}) \setminus \{0\} \longrightarrow E^\times$$

satisfying

$$\begin{aligned} m(2x) &= m(x)^2, \\ m(-x) &= m(x), \end{aligned}$$

$$\prod_{k=0}^{p-1} m(x + kn/p) = m(px) \quad (px \not\equiv 0 \pmod{n}).$$

The true system $m(x) = \bar{\lambda}_x$ is admissible by Lemma 10.1 and Lemma 10.4.

Theorem 10.7 (Freeness). *Choose one representative x_i from each of the g^+ orbits of $\langle 2, -1 \rangle$ on the primitive indices. Evaluation induces a bijection*

$$\{\text{admissible systems}\} \longrightarrow (E^\times)^{g^+}, \quad m \longmapsto (m(x_1), \dots, m(x_{g^+})).$$

Thus the primitive-orbit values may be prescribed arbitrarily and independently.

Proof. Frobenius and sign determine m on each primitive orbit from $m(x_i)$. Every nonzero lower-level index is pz , and the distribution relation determines $m(pz)$ from values one level higher; induction gives injectivity.

Conversely prescribe arbitrary $v_i \in E^\times$ and set $m(\pm 2^k x_i) = v_i^{2^k}$. This is well defined because each orbit has length $\bar{f}$ and $v_i^{2^{\bar{f}}} = v_i$. Define the lower levels recursively by the distribution product. Changing a lift of pz permutes the p factors, and doubling or negating the lift respectively squares or permutes the same product. Hence all three axioms hold. ■

Corollary 10.8 (Multiplicative no-go). *If $g^+ \geq 2$ and x, y are primitive indices in distinct $\langle 2, -1 \rangle$ -orbits, then exactly $(2^{\bar{f}} - 1)^{g^+ - 1}$ admissible systems satisfy $m(x)m(y) = 1$. Therefore the Frobenius, sign, distribution, and norm consequences encoded above cannot by themselves exclude a primitive collision.*

Proof. In the coordinates of Theorem 10.7, prescribe $v_2 = v_1^{-1}$ and choose all remaining coordinates freely. The norm relation is automatic: on each orbit its product is $v_i^{2^{\bar{f}} - 1} = 1$. ■

On a Wieferich plateau $g^+ \geq p^{j-1}$, so this no-go theorem applies precisely where the universal proof is missing. The script `code/ai_prototypes/circular_module.py` constructs explicit systems for $(p, j) = (1093, 2)$ and $(3511, 2)$ that also remain injective on $\{\pm x\}$ -classes, avoid the value 1, and contain a primitive collision. The strengthened construction in `code/ai_prototypes/plateau_endpoint.py` additionally imposes the additive distribution law, the Artin–Schreier trace law, and the global trace law of Theorem 10.24 below. These are formal countermodels to the listed axioms, not counterexamples to the original problem: they deliberately omit the full multiplication identity

$$\bar{\lambda}_x \bar{\lambda}_y = \bar{\lambda}_{x+y} + \bar{\lambda}_{x-y}.$$

They show exactly that any successful proof in this framework must use more of that additive subgroup structure.

10.2.1 Relation weights and Zech logarithms

A positive relation of weight w is a multiset of nonzero $\{\pm x\}$ -classes with product of the corresponding $\bar{\lambda}_x$ equal to 1. The main problem is the assertion that there is no primitive relation of weight two. Within one Frobenius orbit this invariant is explicit. If d_x is the multiplicative order of $\bar{\lambda}_x$, then a relation

$$\prod_k \bar{\lambda}_{2^k x}^{e_k} = 1$$

is equivalent to

$$\sum_k e_k 2^k \equiv 0 \pmod{d_x}.$$

Thus an orbit supports a weight-two relation exactly when $2^i \equiv -1 \pmod{d_x}$ for some i . This recovers the exceptional decomposition-group ratios but says nothing about collisions between different primitive orbits, which are exactly the free coordinates in Theorem 10.7.

Equivalently, if γ generates $\mathbb{F}_{2^f}^\times$, $\theta = \gamma^L$, and Z is the Zech logarithm $1 + \gamma^m = \gamma^{Z(m)}$, then

$$\log_\gamma \bar{\lambda}_x = Z(2xL) - xL,$$

and a collision is

$$Z(2xL) + Z(2yL) \equiv (x + y)L \pmod{2^f - 1}.$$

This is an exact restatement, but available Costas/Sidon properties of Zech logarithms do not control the restriction to two independent p^j -orbits. The freeness theorem explains why orbitwise identities alone cannot do so.

10.3 Ratios in the decomposition group

The decomposition group at $\mathfrak{P}$ is $D = \langle 2 \rangle \subseteq (\mathbb{Z}/n\mathbb{Z})^\times$, and by (23) a ratio $y/x \in D$ is exactly the case in which the two sine residues are Frobenius conjugate. This case can be settled completely and unconditionally.

Theorem 10.9. *Let $x, y \in \mathbb{Z}/n\mathbb{Z}$ be nonzero with $y \equiv 2^i x \pmod{n}$ for some $i \geq 0$, and suppose $\bar{\lambda}_x \bar{\lambda}_y = 1$ and $y \not\equiv \pm x$. Let m be the exact order of θ^x and $f_m = \text{ord}_m(2)$. Then*

$$4 \mid f_m, \quad 2^{f_m/2} \equiv -1 \pmod{m}, \quad y \equiv \pm \rho^{\pm 1} x \text{ with } \rho = 2^{f_m/4}, \quad \rho^2 \equiv -1 \pmod{m},$$

and, writing $\mathbb{F} = \mathbb{F}_{2^{f_m/2}}$ and $\mathbb{F}_0 = \mathbb{F}_{2^{f_m/4}}$,

$$\bar{\lambda}_x \in \mathbb{F}^\times, \quad N_{\mathbb{F}/\mathbb{F}_0}(\bar{\lambda}_x) = 1.$$

Equivalently, in the sum form of Proposition 10.3, $B \equiv \mp \rho A$ and $\text{Tr}_{\mathbb{F}/\mathbb{F}_0}(\bar{\lambda}_A) = 1$.

Proof. Since $y = 2^i x$, both indices have the same p -valuation, so replacing θ by θ^x we may assume $x = 1$ and $n = m$; write $f = f_m$. (This is legitimate: $x(2^i \mp 1) \equiv 0 \pmod{n}$ if and only if $2^i \equiv \pm 1 \pmod{m}$, so the excluded case is the same one, and the conclusion $2^i \equiv \rho^{\pm 1} \pmod{m}$ gives back $y \equiv \rho^{\pm 1} x \pmod{n}$.) By (23) the hypothesis reads $\bar{\lambda}_1^{2^i+1} = 1$, so the multiplicative order of $\bar{\lambda}_1$ divides

$$\gcd(2^i + 1, 2^f - 1) = \begin{cases} 2^d + 1, & f/d \text{ even,} \\ 1, & f/d \text{ odd,} \end{cases} \quad d = \gcd(i, f),$$

a standard identity. If f/d is odd then $\bar{\lambda}_1 = 1$, contradicting Lemma 10.2(2). So f/d is even and $\bar{\lambda}_1 \in \mu_{2^{2d}+1} \subseteq \mathbb{F}_{2^{2d}}^\times$. Then $\bar{\lambda}_{2^{2d}} = \bar{\lambda}_1^{2^{2d}} = \bar{\lambda}_1$, so $2^{2d} \equiv \pm 1 \pmod{n}$ by Lemma 10.2(3).

Suppose $2^{2d} \equiv 1$. Then $f \mid 2d$; since $d \mid f$ and f/d is even, $f = 2d$. Now 2^d is a square root of 1 in the cyclic group $(\mathbb{Z}/n\mathbb{Z})^\times$ and $2^d \neq 1$, so $2^d \equiv -1$. As $d = \gcd(i, f)$ divides i , we get $2^i \in \{2^d, 2^{2d}\} = \{-1, 1\}$, i.e. $y \equiv \pm x$, which was excluded.

Hence $2^{2d} \equiv -1$. Then $f \mid 4d$ but $f \nmid 2d$, and f/d is even, so $f = 4d$; thus $\rho := 2^d = 2^{f/4}$ satisfies $\rho^2 \equiv -1$ and $2^{f/2} \equiv -1$. From $\gcd(i, 4d) = d$ we get $i = kd$ with k odd, so $2^i = \rho^k = \rho^{\pm 1}$. Finally $\bar{\lambda}_1 \in \mu_{2^{2d}+1} \subseteq \mathbb{F}_{2^{2d}}^\times = \mathbb{F}^\times$ and $\bar{\lambda}_1^{2^d+1} = 1$ is exactly $N_{\mathbb{F}/\mathbb{F}_0}(\bar{\lambda}_1) = 1$.

For the sum form, Proposition 10.3 with $y = \rho x$ gives $A = (1 + \rho)x$, $B = (1 - \rho)x$ and $B/A = (1 - \rho)/(1 + \rho) = -\rho$ because $(1 + \rho)^2 = 2\rho$ and $(1 + \rho)(1 - \rho) = 2$. Since $2^{f/2} \equiv -1$ we have $\bar{\lambda}_A^{2^{f/2}} = \bar{\lambda}_{-A} = \bar{\lambda}_A$, so $\bar{\lambda}_A \in \mathbb{F}$, and the relation $\bar{\lambda}_A + \bar{\lambda}_B = \bar{\lambda}_A + \bar{\lambda}_A^{2^{f/4}} = 1$ is $\text{Tr}_{\mathbb{F}/\mathbb{F}_0}(\bar{\lambda}_A) = 1$. $\blacksquare$

Corollary 10.10. *If $-1 \notin \langle 2 \rangle$ in $(\mathbb{Z}/p^j\mathbb{Z})^\times$, or if $4 \nmid \text{ord}_{p^j}(2)$, then no solution of (4) has ratio y/x in the decomposition group $\langle 2 \rangle$. In general the only ratio class in $\langle 2 \rangle$ that can occur is $\{\pm \rho^{\pm 1}\}$ with $\rho = 2^{f/4}$ a primitive fourth root of unity modulo p^j , and in that case the stabilizer of Proposition 9.4 is $H = \{\pm 1, \pm \rho\} \subseteq \langle 2 \rangle$, so its orbits on the primes above 2 are singletons and the reciprocal symmetry gives no lower bound on $|\Sigma_{a,b}|$ beyond 1. (For $p = 17$ one indeed has $|\Sigma| = 1$.)*

Theorem 10.9 disposes of the f ratios forming the decomposition group, against the 8 ratios that cyclotomic units dispose of (Theorem 8.1); the two families always share $r_{\text{prod}} = 2^{\pm 1}$ and generically nothing else. It is sharp: the exceptional class really occurs.

Example 10.11. *For $p = 17$, $f = 8$, $\rho = 2^2 = 4$ and $\rho^2 = 16 \equiv -1$; all eight product-form solutions have ratio in $\{4, 13\} = \{\pm \rho^{\pm 1}\}$, and in fact $|\Sigma| = 1$, matching the example after Proposition 9.4. Moreover the eight sum-form indices of those solutions are exactly the eight A with $\text{Tr}_{\mathbb{F}_{16}/\mathbb{F}_4}(\bar{\lambda}_A) = 1$, as Theorem 10.9 predicts. For $p = 257$, $f = 16$, $\rho = 2^4 = 16$, and of the 144 product-form solutions exactly 16 have ratio in $\langle 2 \rangle$; all of these have ratio $\pm \rho^{\pm 1}$. For $p = 31$ and $p = 127$ one has $-1 \notin \langle 2 \rangle$, and indeed none of the 20, resp. 56, solutions has ratio in $\langle 2 \rangle$. For the Wieferich prime $p = 1093$ and $j = 2$ the exceptional class*

is present ($f = 364$, $\rho = 2^{91}$), so Theorem 10.9 reduces the 364 ratios of the decomposition group to a single class, on which the condition becomes $\text{Tr}_{\mathbb{F}_{2^{182}}/\mathbb{F}_{2^{91}}}(\bar{\lambda}_A) = 1$. For the Wieferich prime $p = 3511$, by contrast, $f = 1755$ is odd, so Corollary 10.10 excludes every ratio in the decomposition group, at every level j of the plateau.

10.4 The ray class of the prime above p as a power-residue condition

We now make the ray-class statement of Section 9 explicit. Recall $\mathfrak{l}_j^+$, the unique prime of K_j^+ above p , and put $\pi_0 = (1 - \zeta_j)(1 - \zeta_j^{-1})$, a generator of $\mathfrak{l}_j^+$, and $\xi_a = s_{j,a}/s_{j,1} \in \mathcal{C}_j^+$ for $p \nmid a$, the standard circular units.

Proposition 10.12. *Let $\mathfrak{q} \mid 2$ in K_j^+ and let a, b be prime to p . Then $s_{j,a}s_{j,b}$ generates $\mathfrak{l}_j^+$ and*

$$s_{j,a}s_{j,b} \equiv 1 \pmod{\mathfrak{q}} \iff \bar{\pi}_0 = (\bar{\xi}_a \bar{\xi}_b)^{-1} \text{ in } (\mathcal{O}_{K_j^+}/\mathfrak{q})^\times.$$

In particular a primitive-primitive collision forces the class of $\mathfrak{l}_j^+$ in the ray class group of K_j^+ of modulus $\mathfrak{q}$ to be trivial, i.e. forces $\mathfrak{l}_j^+$ to split completely in the ray class field of conductor $\mathfrak{q}$; and it forces $\bar{\pi}_0 \in \rho_{\mathfrak{q}}(\mathcal{C}_j^+)$, hence $\bar{\pi}_0 \in \rho_{\mathfrak{q}}(\mathcal{O}_{K_j^+}^\times)$.

Proof. $(s_{j,a}) = \mathfrak{l}_j$ in K_j for $p \nmid a$, so $(s_{j,a}s_{j,b}) = \mathfrak{l}_j^2 = \mathfrak{l}_j^+ \mathcal{O}_{K_j}$, and $s_{j,a}s_{j,b}$ is fixed by complex conjugation because $s_{j,-a} = -s_{j,a}$. Also $s_{j,1}^2 = -\pi_0$, so $s_{j,a}s_{j,b} = -\pi_0 \xi_a \xi_b$. Reducing modulo $\mathfrak{q}$ and using $-1 \equiv 1$ gives the displayed equivalence. The ray-class statement is the exact sequence $1 \rightarrow (\mathcal{O}/\mathfrak{q})^\times / \rho_{\mathfrak{q}}(\mathcal{O}^\times) \rightarrow Cl_{\mathfrak{q}} \rightarrow Cl \rightarrow 1$ together with the fact that the class of the principal ideal (π_0) in $Cl_{\mathfrak{q}}$ is the class of $\bar{\pi}_0$. $\blacksquare$

Because $(\mathcal{O}_{K_j^+}/\mathfrak{q})^\times$ is cyclic of odd order, a membership statement in it is equivalent to the vanishing of finitely many power-residue symbols. This is the concrete form of the “semi-local units modulo circular units” condition.

Definition 10.13. *For a prime $\ell \mid 2^f - 1$ let $\chi_\ell : \mathbb{F}_q^\times \rightarrow \mu_{\ell^{C_\ell}}$, $\chi_\ell(z) = z^{(q-1)/\ell^{C_\ell}}$, with $C_\ell = v_\ell(q-1)$, be the ℓ -primary power-residue symbol, fix a generator ξ of $\mu_{\ell^{C_\ell}}$ and define $\Psi_\ell : \mathbb{Z}/n\mathbb{Z} \setminus \{0\} \rightarrow \mathbb{Z}/\ell^{C_\ell}\mathbb{Z}$ by $\chi_\ell(\bar{\lambda}_x) = \xi^{\Psi_\ell(x)}$.*

Lemma 10.14. $\Psi_\ell(2x) = 2\Psi_\ell(x)$, $\Psi_\ell(-x) = \Psi_\ell(x)$, $\sum_{t \in (\mathbb{Z}/n\mathbb{Z})^\times} \Psi_\ell(t) = 0$, and $\sum_{t \in H_\beta} \Psi_\ell(xt) = \Psi_\ell(p^\beta x)$. A solution of (4) with product-form indices x, y forces

$$\Psi_\ell(x) + \Psi_\ell(y) \equiv 0 \pmod{\ell^{C_\ell}} \quad \text{for every prime } \ell \mid 2^f - 1, \quad (24)$$

and conversely (24) for all ℓ is equivalent to $\bar{\lambda}_x \bar{\lambda}_y = 1$.

Proof. The first two rules are (23), the last two are Lemma 10.4, and (24) is χ_ℓ applied to $\bar{\lambda}_x \bar{\lambda}_y = 1$. For the converse, $\mathbb{F}_q^\times$ is cyclic of order $q-1$ and the χ_ℓ jointly separate its elements. $\blacksquare$

10.5 The p -primary symbol is determined by Stickelberger eigenspaces

The member $\ell = p$ of the family in Lemma 10.14 is the one carrying arithmetic: $p^C \parallel q-1$ with $C = v_p(q-1) = \max(c_p, j)$, and χ_p is the p -power residue symbol whose Gauss and Jacobi sums are governed by Stickelberger’s theorem. We determine it completely. Write $\Psi = \Psi_p$, $G = (\mathbb{Z}/n\mathbb{Z})^\times = G' \times G_p$ with G' cyclic of order $p-1$ and G_p the principal units of order p^{j-1} , let $\tau : G' \rightarrow \mu_{p-1} \subseteq (\mathbb{Z}/p^C\mathbb{Z})^\times$ be the Teichmüller isomorphism, and let $f_1 = \text{ord}_p(2)$.

Theorem 10.15. *Let $A = \{k \bmod p-1 : k \text{ even, } k \equiv 1 \pmod{f_1}\}$. Then:*

1. $\Psi = \sum_{k \in A} \Psi_k$, where Ψ_k is the τ^k -eigencomponent, i.e. $\Psi_k(tx) = \tau(t)^k \Psi_k(x)$ for $t \in G'$. In particular $|A| = (p-1)/(2f_1)$ if f_1 is odd, and $A = \emptyset$ if f_1 is even.
2. If f_1 is even then $\Psi \equiv 0$: every $\bar{\lambda}_x$ is a p^C -th power in $\mathbb{F}_q^\times$, and the whole p -primary obstruction is vacuous.
3. $\Psi \bmod p$ is invariant under the subgroup $\langle v \rangle \subseteq G_p$ of order $p^{\max(0, j-c_p)}$ generated by the p -part v of the element $2 \in G$; equivalently, $\Psi(x) \bmod p$ depends only on x modulo $p^{\min(j, c_p)}$.
4. If $1 \leq \beta \leq j - c_p$ then $\Psi(p^\beta x) \equiv 0 \pmod{p}$.

Proof. Let $M = \text{Fun}(G, \mathbb{Z}/p^C\mathbb{Z})$ with the translation action $(t\Psi)(x) = \Psi(tx)$. Since $|G'| = p-1$ is invertible in $\mathbb{Z}/p^C\mathbb{Z}$ and $\mu_{p-1} \subseteq (\mathbb{Z}/p^C\mathbb{Z})^\times$ by Hensel's lemma, the idempotents $e_k = (p-1)^{-1} \sum_{t \in G'} \tau(t)^{-k} t$ decompose $M = \bigoplus_{k \bmod p-1} M_k$, and each M_k is a module over the local ring $\Lambda = \mathbb{Z}/p^C\mathbb{Z}[G_p]$ with maximal ideal $\mathfrak{m} = (p, g-1 : g \in G_p)$ and residue field $\mathbb{F}_p$.

By Lemma 10.14, Ψ satisfies $\sigma_{-1}\Psi = \Psi$ and $\sigma_2\Psi = 2\Psi$, where σ_t is translation by the group element t and the right-hand 2 is the scalar $2 \in \mathbb{Z}/p^C\mathbb{Z}$. As $\tau(-1) = -1$, the element σ_{-1} acts on M_k by $(-1)^k$, so $\Psi_k = 0$ for odd k . Write $2 = \omega v$ with $\omega \in G'$, $v \in G_p$; then $\omega \equiv 2 \pmod{p}$, so σ_ω acts on M_k by $\tau(2)^k$ and the condition reads

$$(\sigma_v - u_k) \Psi_k = 0 \text{ in } M_k, \quad u_k = 2\tau(2)^{-k} \in (\mathbb{Z}/p^C\mathbb{Z})^\times.$$

Since $\sigma_v \equiv 1 \pmod{\mathfrak{m}}$, the element $\sigma_v - u_k$ is congruent to the scalar $1 - u_k$ modulo $\mathfrak{m}$, hence is a unit of Λ unless $u_k \equiv 1 \pmod{p}$. Now $u_k \equiv 2^{1-k} \pmod{p}$, so $u_k \equiv 1$ if and only if $k \equiv 1 \pmod{f_1}$. This proves (1); the count of A is the number of even residues in the arithmetic progression $1 + f_1\mathbb{Z}$ modulo $p-1$, which is $(p-1)/(2f_1)$ when f_1 is odd and 0 when f_1 is even. Statement (2) is the case $A = \emptyset$.

For (3), let $k \in A$. Then $\tau(2)^k = \tau(2)$ because $\tau(2)$ has order f_1 , so $u_k = 2/\tau(2) = \langle 2 \rangle$ is the principal unit part of 2, and

$$v_p(\langle 2 \rangle - 1) = v_p(\langle 2 \rangle^{p-1} - 1) = v_p(2^{p-1} - 1) = c_p,$$

using that $\langle 2 \rangle \in 1 + p\mathbb{Z}_p$ and that $p-1$ is prime to p . Hence $u_k \equiv 1 \pmod{p}$ and the relation $(\sigma_v - u_k)\Psi_k = 0$ gives $(\sigma_v - 1)\Psi_k \equiv 0 \pmod{p}$, i.e. $\Psi_k \bmod p$ is v -invariant. Since $\text{ord}_G(2) = f = f_1 p^{\max(0, j-c_p)}$ and ω has order f_1 prime to p , the element v has order $p^{\max(0, j-c_p)}$, and $\langle v \rangle$ is the kernel of the reduction $(\mathbb{Z}/p^j\mathbb{Z})^\times \rightarrow (\mathbb{Z}/p^{\min(j, c_p)}\mathbb{Z})^\times$, being the unique subgroup of that order in the cyclic group G_p .

For (4), apply Lemma 10.14 with $H_\beta \subseteq \langle v \rangle$, which holds for $\beta \leq j - c_p$: modulo p the left-hand side is $p^\beta \Psi(x) \equiv 0$. ■

Part (2) is already decisive for one of the two known Wieferich primes.

Corollary 10.16. *For $p = 1093$ one has $\text{ord}_p(2) = 364$, which is even, so $\Psi \equiv 0$ at every level j : every $\bar{\lambda}_x$ is a p^C -th power in $\mathbb{F}_{2^{364}}^\times$. No Stickelberger, Jacobi-sum or p -power-residue argument can contribute anything at all to the case $p = 1093$.*

When f_1 is odd the symbol is nonzero in every case we have computed, and in the extreme case $|A| = 1$ it has a closed form, which yields a theorem conditional only on the single nonvanishing $\alpha \neq 0$.

Theorem 10.17. *Let $p \equiv 7 \pmod{8}$ with $\text{ord}_p(2) = (p-1)/2$, so that f_1 is odd and $A = \{k_0\}$ with $k_0 = f_1 + 1$. Let $j \geq 1$ satisfy $\min(j, c_p) = 1$, i.e. $j = 1$, or p is not a Wieferich prime. Assume $\alpha := \Psi(1) \not\equiv 0 \pmod{p}$, i.e. $\bar{\lambda}_1$ is not a p -th power in $\mathbb{F}_q^\times$. Then*

$$\Psi(x) \equiv \alpha \bar{x} \left(\frac{\bar{x}}{p} \right) \pmod{p} \quad (p \nmid x), \quad \Psi(x) \equiv 0 \pmod{p} \quad (p \mid x),$$

where $\bar{x} = x \bmod p$ and $\left(\frac{\cdot}{p} \right)$ is the Legendre symbol. Moreover every pair x, y with $\bar{\lambda}_x \bar{\lambda}_y = 1$ has $p \mid x$ and $p \mid y$; that is, Problem 1.1 has an affirmative answer for (p, j) .

Proof. The residues modulo $p - 1 = 2f_1$ that are $\equiv 1 \pmod{f_1}$ are 1 and $1 + f_1$; as f_1 is odd, exactly the second is even, so $A = \{k_0\}$, $k_0 = f_1 + 1 = (p + 1)/2$. By Theorem 10.15(1) and (3), and because $\min(j, c_p) = 1$, the function $\Psi \bmod p$ factors through $(\mathbb{Z}/p\mathbb{Z})^\times$: writing $x = tu$ with $t \in G'$ and $u \in G_p$ one has $\Psi(x) = \tau(t)^{k_0} \Psi(u)$ and $\Psi(u) \equiv \Psi(1) \bmod p$, because $u \equiv 1 \pmod{p}$; since $\tau(t) \equiv t \equiv x \pmod{p}$ this gives $\Psi(x) \equiv \alpha \bar{x}^{k_0}$. Euler's criterion gives $\bar{x}^{k_0} = \bar{x}^{(p+1)/2} = \bar{x} \cdot \bar{x}^{(p-1)/2} = \bar{x} \left(\frac{\bar{x}}{p} \right)$. Theorem 10.15(4) with $c_p = 1$ gives $\Psi(p^\beta x) \equiv 0$ for $\beta \geq 1$.

Put $F(x) = \bar{x} \left(\frac{\bar{x}}{p} \right)$. If $\bar{x}$ is a quadratic residue then $F(x) = \bar{x}$ is a quadratic residue; if $\bar{x}$ is a non-residue then $F(x) = -\bar{x}$, again a quadratic residue because $p \equiv 3 \pmod{4}$ makes -1 a non-residue. So F takes only nonzero square values.

Suppose $\bar{\lambda}_x \bar{\lambda}_y = 1$ with $x, y \neq 0$ and, after interchanging x and y , with $p \nmid x$. By Lemma 10.14, $\Psi(x) + \Psi(y) \equiv 0 \pmod{p}$. If $p \mid y$ this reads $\alpha F(x) \equiv 0$, impossible since $\alpha \neq 0$ and $F(x) \neq 0$. If $p \nmid y$ it reads $F(x) = -F(y)$, with $F(x)$ a square and $-F(y)$ a non-zero non-square: impossible. Thus every solution has $p \mid x$ and $p \mid y$, which by Proposition 10.3 is the assertion of Problem 1.1. $\blacksquare$

Corollary 10.18. *Under the hypotheses of Theorem 10.17 for every level $1 \leq j' \leq j$, one has $S_{p^{j'}} = \emptyset$ and $d(p^j - 1) = 0$.*

Proof. Induct on j . For $j = 1$ every nonzero index is prime to p , so Theorem 10.17 gives $S_p = \emptyset$. For $j > 1$, Theorem 10.17 leaves only pairs with $p \mid x$ and $p \mid y$; such a pair is a solution for θ^p , of exact order p^{j-1} , hence is excluded by the inductive hypothesis. Now apply Lemma 1.4. $\blacksquare$

Corollary 10.19. *For $p = 3511$ one has $p \equiv 7 \pmod{8}$ and $\text{ord}_p(2) = 1755 = (p - 1)/2$, and $\bar{\lambda}_1$ is not a p -th power in $\mathbb{F}_{2^{1755}}^\times$ (verified in `code/ai_prototypes/sine_unit_obstruction.py`). Hence $S_{3511} = \emptyset$ and $d(3510) = 0$, by a power-residue argument using one power-residue computation but no enumeration of exponent pairs. The same conclusion is obtained unconditionally, without a power-residue or finite-field computation, in Corollary 10.23 below.*

Example 10.20. *The hypotheses of Theorem 10.17 hold for $p = 7, 23, 47, 71, 79, 103, 167, 191, 199, 239, \dots$, and in every one of these cases the computation confirms both $\alpha \neq 0$ and the closed form $\Psi(x) \equiv \alpha \bar{x} \left(\frac{\bar{x}}{p} \right)$ at level $j = 1$, hence $d(p - 1) = 0$; the same was confirmed at $j = 2, 3$ for $p = 7$ and at $j = 2$ for $p = 23$, so Corollary 10.18 gives $d(7^j - 1) = 0$ for $j \leq 3$ and $d(23^j - 1) = 0$ for $j \leq 2$. For $p = 7$ this reproves, in three lines, the case $p^j = 49$ that Table 3 settles by fourteen resultant factorizations. The obstruction is not always this strong: for $p = 31$ and $p = 127$, where $|A| = 3$ and $|A| = 9$, the admissible ratio classes cut out by (24) with $\ell = p$ are exactly the ratio classes of the solutions that do exist, so the p -part is sharp but not decisive; and for $p = 17$, where $f_1 = 8$ is even, it is empty.*

10.6 The trace obstruction: an unconditional replacement

The hypothesis $\alpha \neq 0$ of Theorem 10.17 can be removed entirely, at the cost of replacing the multiplicative power-residue symbol by an additive one. Let $\bar{f} = \rho_2(p^j)$, let $E = \mathbb{F}_{2^{\bar{f}}}$ be the residue field of the prime $\mathfrak{q} \mid 2$ of K_j^+ — so that $\bar{\lambda}_x \in E$ for every x — and put

$$\delta = \begin{cases} 1, & -1 \in \langle 2 \rangle \subseteq (\mathbb{Z}/p^j\mathbb{Z})^\times \quad (\text{equivalently } f = 2\bar{f}), \\ 0, & \text{otherwise,} \end{cases} \quad \tau(x) = \text{Tr}_{E/\mathbb{F}_2}(\bar{\lambda}_x).$$

Theorem 10.21. *For every $x \not\equiv 0 \pmod{p^j}$ one has $\text{Tr}_{E/\mathbb{F}_2}(\bar{\lambda}_x^{-1}) = \delta$. Consequently, if $\bar{\lambda}_x \bar{\lambda}_y = 1$ then*

$$\tau(x) = \tau(y) = \delta.$$

Proof. Put $u = \theta^x$ and $c = \bar{\lambda}_x$, so $u^2 + cu + 1 = 0$; substituting $u = cS$ gives the Artin–Schreier equation $S^2 + S = c^{-2}$, whose roots are u/c and u^{-1}/c . It has a root in E if and only if $\text{Tr}_{E/\mathbb{F}_2}(c^{-2}) =$

$\text{Tr}_{E/\mathbb{F}_2}(c^{-1}) = 0$, and also if and only if $u \in E$. Now $u \in E$ exactly when $2^{\bar{f}} \equiv 1$, i.e. when $\delta = 0$: for $\delta = 1$ one has $u^{2^{\bar{f}}} = u^{-1} \neq u$ because $u^2 \neq 1$. For the consequence, $\bar{\lambda}_y = \bar{\lambda}_x^{-1}$, so $\tau(x) = \text{Tr}_{E/\mathbb{F}_2}(\bar{\lambda}_y^{-1}) = \delta$ and symmetrically. ■

Theorem 10.22. *Suppose $\text{ord}_p(2) = \varphi(p^j)/2$ is odd, i.e. that $\langle 2 \rangle$ has index two in $(\mathbb{Z}/p^j\mathbb{Z})^\times$ and $-1 \notin \langle 2 \rangle$. Then $\delta = 0$, $E = \mathbb{F}_{2^f}$, and*

$$\tau(x) = \begin{cases} 1, & v_p(x) = j-1, \\ 0, & v_p(x) \leq j-2. \end{cases}$$

Hence every solution of (4) has $v_p(x), v_p(y) \leq j-2$ in product form; for $j = 1$ there is none, so $S_p = \emptyset$ and $d(p-1) = 0$.

Proof. Oddness of $\varphi(p^j)/2 = p^{j-1}(p-1)/2$ forces $p \equiv 3 \pmod{4}$, so -1 is a non-square modulo every power of p ; and $\langle 2 \rangle$, being of index two in a cyclic group, is the group of squares. Hence $\delta = 0$, $\bar{f} = f$ and $\theta \in E$. Put $t(x) = \text{Tr}_{E/\mathbb{F}_2}(\theta^x)$, so $\tau(x) = t(x) + t(-x)$ and $t(2x) = t(x)$.

Fix $\beta \leq j-1$. Reduction $(\mathbb{Z}/p^j\mathbb{Z})^\times \rightarrow (\mathbb{Z}/p^{j-\beta}\mathbb{Z})^\times$ carries squares onto squares, so the image of $\langle 2 \rangle$ again has index two and misses -1 ; therefore the indices of level β split into exactly two $\langle 2 \rangle$ -orbits, interchanged by $x \mapsto -x$, and t is constant on each, with values $t_\beta^\pm$. Thus τ is constant on level β , equal to $\tau_\beta = t_\beta^+ + t_\beta^-$.

Summing t over level β and using that in characteristic two the sum of all p^k -th roots of unity vanishes for $k \geq 1$, so that the sum of the primitive p^k -th roots of unity is 0 for $k \geq 2$ and 1 for $k = 1$,

$$\frac{\varphi(p^{j-\beta})}{2} \tau_\beta = \sum_{v_p(x)=\beta} t(x) = \begin{cases} \text{Tr}_{E/\mathbb{F}_2}(0) = 0, & \beta \leq j-2, \\ \text{Tr}_{E/\mathbb{F}_2}(1) = f \bmod 2 = 1, & \beta = j-1. \end{cases}$$

Since $\varphi(p^{j-\beta})/2$ is odd, reduction modulo 2 gives $\tau_\beta = 0$ for $\beta \leq j-2$ and $\tau_{j-1} = 1$. Now apply Theorem 10.21: a collision needs $\tau(x) = \tau(y) = \delta = 0$. Finally $d(p-1) = |S_p|/2$ by Lemma 1.4. ■

Corollary 10.23. *Let $p \equiv 7 \pmod{8}$ with $\text{ord}_p(2) = (p-1)/2$. Then $S_p = \emptyset$ and $d(p-1) = 0$ unconditionally: no power-residue nonvanishing hypothesis is needed. In particular $d(3510) = 0$ is a theorem, and the same holds for $p = 7, 23, 47, 71, 79, 103, 167, 191, 199, \dots$*

Proof. The hypothesis gives $\text{ord}_p(2) = \varphi(p)/2$, odd because $p \equiv 3 \pmod{4}$. Apply Theorem 10.22 with $j = 1$. ■

10.7 A global trace law

The Artin-Schreier obstruction uses the trace of $\bar{\lambda}_x$. A different trace identity appears after rewriting the collision as a weight-five “box plus centre” relation.

Theorem 10.24 (Global trace law). *Let*

$$T(x) = \text{Tr}_{\mathbb{F}_{2^f}/\mathbb{F}_2}(\theta^x).$$

If $\bar{\lambda}_x \bar{\lambda}_y = 1$, then

$$T(x) + T(y) = f \pmod{2}. \quad (25)$$

Proof. Multiplying $(\theta^x + \theta^{-x})(\theta^y + \theta^{-y}) = 1$ by θ^{x+y} gives

$$1 + \theta^{2x} + \theta^{2y} + \theta^{2x+2y} + \theta^{x+y} = 0.$$

Apply the absolute trace. Frobenius invariance gives $\text{Tr}(\theta^{2x}) = T(x)$, $\text{Tr}(\theta^{2y}) = T(y)$, and $\text{Tr}(\theta^{2x+2y}) = T(x+y)$; the last term contributes another $T(x+y)$ and cancels it. Finally $\text{Tr}(1) = f \bmod 2$. ■

Corollary 10.25 (Three exact new cases). *Exact trace enumeration gives*

$$d(72) = d(88) = d(232) = 0,$$

corresponding to $p = 73, 89, 233$.

Proof. For each prime, enumerate the indices satisfying the necessary Artin–Schreier condition of Theorem 10.21, split them by $T(x)$, and apply (25). In all three cases no ordered pair survives both conditions. The computation is exact bit-vector arithmetic in `code/ai_prototypes/plateau_endpoint.py`; direct evaluation of S_p independently confirms the empty set. ■

Further applications of the box-plus-centre identity produce additional linear trace relations, for example

$$T(x - 2y) + T(2x - y) + T(3x - 3y) = 0.$$

They strengthen the filter in individual cases but have not yielded a uniform endpoint theorem. The formal constructions following Corollary 10.8 can be chosen to satisfy the additive distribution law, Theorem 10.21, Theorem 10.24, and the known level sums while still containing a primitive collision at both known Wieferich plateaus. Thus these trace laws are genuine progress, but any general proof must use the full nonlinear multiplication law rather than only the currently extracted linear consequences.

10.8 Exact Kloosterman formulas at the full-torus extremes

For $r \geq 1$ define the binary Kloosterman sum

$$K_r = \sum_{z \in \mathbb{F}_{2^r}} (-1)^{\text{Tr}_{\mathbb{F}_{2^r}/\mathbb{F}_2}(z+z^{-1})}, \quad 0^{-1} = 0.$$

Theorem 10.26 (Mersenne and Fermat formulas). *1. If $p = 2^r - 1$ is a Mersenne prime, $r \geq 3$, then*

$$d(p-1) = \frac{p-3+K_r}{2}.$$

2. If $p = 2^m + 1$ is a Fermat prime, $m \geq 2$, then

$$d(p-1) = \frac{p-1+K_m}{2}.$$

Proof. In the Mersenne case $\mu_p = \mathbb{F}_{2^r}^\times$; in the Fermat case μ_p is the full norm-one torus in $\mathbb{F}_{2^{2m}}/\mathbb{F}_{2^m}$. Therefore the trace-value set $\{\lambda_x : x \neq 0\}$ is the entire Artin–Schreier set described by Theorem 10.21, so its necessary condition is also sufficient.

Expand the two trace indicators for a pair $\alpha, \alpha+1$ and sum over $\alpha \neq 0, 1$. The linear terms are evaluated by orthogonality of the additive character. For the quadratic term use

$$\alpha^{-1} + (\alpha+1)^{-1} = (\alpha^2 + \alpha)^{-1}$$

and the two-to-one map $\alpha \mapsto \alpha^2 + \alpha$ onto the trace-zero subspace. The remaining character sum is exactly K_r or K_m . Finally use Lemma 1.4. This gives the two formulas. ■

By the Weil bound, and more sharply by the congruences and recurrences for binary Kloosterman sums [LW90, Car69], $d(p-1) > 0$ for every Mersenne prime $p \geq 31$ and every Fermat prime $p \geq 5$. This explains the level-one solutions for $p = 17, 31, 127, 257$. When μ_{p^j} is a proper subgroup of the torus, the same expansion becomes an average of twisted Kloosterman sums. The Weil error then exceeds the expected main term on the plateau, so the exact full-torus evaluation does not extend to the endpoint problem.

10.9 Why the method stops exactly at the plateau

Proposition 10.27. *Let $2 \leq j \leq c_p$, so that (p, j) lies on the plateau. Then in the notation of Theorem 10.15, $v = 1$ and $C = c_p$, so $u_k = \langle 2 \rangle \equiv 1 \pmod{p^C}$ and the relation $(\sigma_v - u_k)\Psi_k = 0$ is the trivial relation $0 = 0$. Beyond the eigenspace support of Theorem 10.15(1) and the norm relation of Lemma 10.4, no constraint on Ψ survives; the ambient eigenspace has $p^{j-1} - 1$ free parameters in $\mathbb{Z}/p^C\mathbb{Z}$ after fixing its augmentation.*

Proof. On the plateau $\text{ord}_{p^j}(2) = \text{ord}_p(2) = f_1$ is prime to p , so the p -part v of $2 \in G$ is trivial, and $C = v_p(2^f - 1) = c_p \geq j$. By the computation in the proof of Theorem 10.15(3), $v_p(u_k - 1) = c_p = C$, i.e. $u_k = 1$ in $\mathbb{Z}/p^C\mathbb{Z}$; so the relation imposed by σ_2 is vacuous. The eigenspace component M_k is free of rank p^{j-1} over $\mathbb{Z}/p^C\mathbb{Z}$ and the relation of Lemma 10.4 fixes only its augmentation. ■

Heuristic 10.28. *If the remaining free values of Ψ behave randomly, then the necessary condition (24) for $\ell = p$ should remove a proportion about $1 - p^{-c_p}$ of pairs, leaving on the order of $\varphi(p^j)^2 p^{-c_p}$ candidates. At $j = c_p$ this is on the order of p^j . This explains why the p -primary condition is not expected to decide Problem 1.1, but it is not a count of the actual values of Ψ .*

Example 10.29 (the plateau case $p = 3511$, $j = 2$). Here $f_1 = 1755$ is odd, $c_p = 2$, $A = \{1756\}$, and Theorem 10.17 proves the case $j = 1$ outright. At $j = 2$ the descent of Theorem 10.15(3) says only that $\Psi \bmod p$ depends on x modulo p^2 , which is no information, and the p values of Ψ on G_p are unconstrained. The computation `code/ai_prototypes/plateau_ppart_3511.py` evaluates Ψ on 140 elements of G_p inside $\mathbb{F}_{2^{1755}}$, confirms the eigenrelation $\chi(\bar{\lambda}_{tx}) = \chi(\bar{\lambda}_x)^{\tau(t)^{k_0}}$, and finds 4 sampled pairs (u_1, u_2) satisfying the p -primary necessary condition (24) — against 2.79 expected at random. This is heuristic evidence that the obstruction, decisive at level 1, does not decide level 2; these are not five-term solutions.

Remark 10.30. *The same eigenspace analysis applies to $\ell \neq p$: Ψ_ℓ is supported on those characters ψ of G with values in $\mathbb{F}_\ell$ that satisfy $\psi(2) = 2$ and $\psi(-1) = 1$. Such characters exist for many ℓ — the constraint $\psi(2) = 2$ alone is always solvable, because $\text{ord}_\ell(2)$ divides $\text{ord}_G(2) = f$ — so no analogue of Theorem 10.15(2) is available in general. Moreover, if $2^i \equiv -1 \pmod{\ell^{C_\ell}}$ for some i , then every pair $(x, 2^i x)$ satisfies (24) for that ℓ , so the ℓ -primary condition alone can never be empty. Those pairs are precisely the ones removed by Theorem 10.9; the two obstructions of this section are therefore complementary, and neither is subsumed by the other. Individual $\ell \neq p$ can nevertheless be decisive: for $p = 7$, $j = 2$ each of $\ell = 7, 127, 337$ already excludes every admissible pair, and for $p = 23$ both $\ell = 23$ and $\ell = 89$ do. This is not a usable route on the plateau, since it presupposes the factorization of $2^{364} - 1$ or $2^{1755} - 1$.*

Remark 10.31. *Theorem 10.17 is genuinely characteristic-two input, and is therefore not in conflict with Corollary 13.2. Replacing 2 by another prime ℓ , the Frobenius relation becomes $\bar{\lambda}_{\ell x} = \bar{\lambda}_x^\ell$ and the eigenspace condition of Theorem 10.15 becomes $k \equiv 1 \pmod{\text{ord}_p(\ell)}$, so the admissible set A has $(p-1)/(2\text{ord}_p(\ell))$ elements and the argument degrades as $\text{ord}_p(\ell)$ shrinks. For the counterexample $\ell = 491$, $p = 7$, $j = 2$ of Table 1 one has $\text{ord}_7(491) = 1$, so A consists of all even k , the symbol is unconstrained, and no contradiction arises — exactly as it must be.*

11 Modular vanishing sums: what the theorems give

11.1 Congruence modulo the rational prime 2

Theorem 11.1 (Dvornicich–Zannier [DZ02]). *Let ℓ be a rational prime, let $\zeta_0 = 1, \zeta_1, \dots, \zeta_{k-1}$ be pairwise distinct roots of unity, and let $a_i \in \mathbb{Q}^\times$ satisfy $\sum_i a_i \zeta_i \equiv 0 \pmod{\ell}$ while no proper subsum is $\equiv 0 \pmod{\ell}$. Then $Q = \text{lcm}_i \text{ord}(\zeta_i)$ is squarefree and*

$$\sum_{q|Q} (q-2) \leq k-2,$$

the sum being over primes q dividing Q .

We flag explicitly that the congruence in Theorem 11.1 is modulo the rational prime ℓ , that is, in $\mathbb{Z}[\zeta]/\ell\mathbb{Z}[\zeta]$, and not modulo a prime ideal above ℓ . This is not a technicality.

Proposition 11.2. *The literal analogue of Theorem 11.1 for congruences modulo a prime ideal is false. In $\mathbb{F}_8 = \mathbb{F}_2[x]/(x^3 + x + 1)$ the element $\alpha = x$ has order 7 and $\alpha^3 = \alpha + 1$, so $1 + \alpha + \alpha^3 = 0$. Lifting, there is a prime $\mathfrak{p} \mid 2$ of $\mathbb{Z}[\zeta_7]$ with*

$$1 + \zeta_7 + \zeta_7^3 \equiv 0 \pmod{\mathfrak{p}},$$

and no proper subsum is $\equiv 0 \pmod{\mathfrak{p}}$, because $1 + \zeta_7^e$ has norm $\Phi_7(-1) = 1$ and is therefore a unit for $e \neq 0$. Here $k = 3$ and $Q = 7$, so the conclusion of Theorem 11.1 would read $5 \leq 1$.

Nevertheless the congruence modulo the rational prime does apply to our situation, and yields the following. We give an elementary self-contained proof, which is stronger in that it requires no hypothesis on subsums.

Theorem 11.3. *Let $p > 5$ be prime and $j \geq 1$. Then for all a, b ,*

$$\Lambda_{a,b} \not\equiv 0 \pmod{2\mathbb{Z}[\zeta_{p^j}]}.$$

Equivalently, $\Phi_{p^j}(x)$ does not divide $1 + x^a + x^{-a} + x^b + x^{-b}$ in $\mathbb{F}_2[x]/(x^{p^j} + 1)$; that is, (4) can never hold simultaneously for all primitive p^j -th roots of unity in $\overline{\mathbb{F}_2}$.

Proof. Work in $R = \mathbb{F}_2[x]/(x^{p^j} + 1)$. Since $x^{p^j} + 1 = \Phi_{p^j}(x) \cdot (x^{p^{j-1}} + 1)$, every element of the ideal $\Phi_{p^j}R$ has a unique representative $\Phi_{p^j}(x)A(x)$ with $\deg A < p^{j-1}$. Now $\Phi_{p^j}(x) = \Phi_p(x^{p^{j-1}}) = \sum_{k=0}^{p-1} x^{kp^{j-1}}$, so

$$\Phi_{p^j}(x)A(x) = \sum_{k=0}^{p-1} x^{kp^{j-1}} A(x),$$

and the p summands have pairwise disjoint supports, contained in the intervals $[kp^{j-1}, (k+1)p^{j-1})$. Hence

$$\text{wt}(\Phi_{p^j}A) = p \cdot \text{wt}(A),$$

so every nonzero element of $\Phi_{p^j}R$ has weight a positive multiple of p , in particular weight at least p .

The image of $1 + x^a + x^{-a} + x^b + x^{-b}$ in R has weight at most 5, and its coefficient at the exponent 0 equals $1 + 2[a \equiv 0] + 2[b \equiv 0] = 1$ in $\mathbb{F}_2$, so it is nonzero. Since $5 < p$, it is not in $\Phi_{p^j}R$. $\blacksquare$

Remark 11.4. *Theorem 11.1 yields the same conclusion, at the cost of verifying its hypotheses. Assume $\Lambda_{a,b} \equiv 0 \pmod{2}$. By Lemma 1.2 (whose proof applies verbatim modulo 2, using that $1 + \zeta^e$ is a unit for $e \neq 0$) the five roots are distinct. No subsum of size 1 or 2 is $\equiv 0 \pmod{2}$, again because $1 + \zeta^e$ is a unit. For a subsum of size $k \in \{3, 4\}$ with no vanishing proper subsum, Theorem 11.1 forces its conductor $Q \mid p^j$ to be squarefree with $\sum_{q \mid Q} (q - 2) \leq k - 2 \leq 2$, so $Q = 1$ and all its roots equal 1, contradicting distinctness. Applying Theorem 11.1 to the full five-term sum gives Q squarefree, hence $Q = p$, and $p - 2 \leq 3$, i.e. $p \leq 5$. The elementary weight argument is preferable because it is unconditional on the exact wording of the published theorem, which we were unable to consult directly, and because it gives the sharper statement that all weights in $\Phi_{p^j}R$ are multiples of p .*

11.2 Why no such theorem can hold for a single prime ideal

Proposition 11.5. *Let $\theta \in \mathbb{F}_{2^f}$ have exact order N , $f = \text{ord}_N(2)$, and let w_0 be minimal with $\sum_{i=0}^{w_0} \binom{N}{i} > 2^f$. Then there are $w \leq 2w_0$ pairwise distinct residues $k_1, \dots, k_w$ modulo N with $\theta^{k_1} + \dots + \theta^{k_w} = 0$.*

Proof. The $\mathbb{F}_2$ -linear map $\mathbb{F}_2^{\mathbb{Z}/N\mathbb{Z}} \rightarrow \mathbb{F}_{2^f}$ sending $\delta \mapsto \sum_k \delta_k \theta^k$ is surjective. By the pigeonhole principle two distinct vectors of weight at most w_0 have the same image; their difference is a nonzero kernel vector of weight at most $2w_0$. ■

Proposition 11.5 shows that modular vanishing sums of small weight are unavoidable as soon as N is large compared with 2^f ; no Mann-type or Conway–Jones-type inequality can survive reduction modulo a single prime ideal. In coding-theoretic language [MS77, Ch. 7], the kernel in Proposition 11.5 is the dual of the irreducible cyclic code of length N and dimension f generated by the sequence $\left(\mathrm{Tr}_{\mathbb{F}_{2^f}/\mathbb{F}_2}(\lambda \theta^k)\right)_k$, and the statement is the Gilbert–Varshamov-type upper bound on its minimum distance; Proposition 11.2 is the observation that this distance can be as small as 3. Lam and Leung reached the same conclusion from the opposite direction: they determined the weight sets $W_\ell(m)$ of vanishing sums of m -th roots of unity in $\overline{\mathbb{F}}_\ell$ and observed that the characteristic-zero answer fails, remarking that they are “left with no viable conjecture on the structure of the weight set $W_p(m)$ in characteristic p ” [LL96, §1].

11.3 The exact size of the gap

Theorem 11.6. *Let $p > 5$ and $2 \leq j \leq c_p$, so that j lies on the plateau and $f = \mathrm{ord}_{p^j}(2) = \mathrm{ord}_p(2)$ divides $p - 1$. Suppose (a, b) is a counterexample to Problem 1.1, with $\gcd(a, p) = 1$ after possibly interchanging a and b . Then*

$$1 \leq |\Sigma_{a,b}| \leq g - 1, \quad g = \frac{\varphi(p^j)}{f} \geq p^{j-1} \geq p > 5,$$

and $|\Sigma_{a,b}| \geq 2$ whenever $-1 \notin \langle 2 \rangle$ in $(\mathbb{Z}/p^j\mathbb{Z})^\times$. Proposition 9.4 cannot give a lower bound exceeding $|H| \leq 4$.

Proof. The lower bounds are Proposition 9.4 together with $\Sigma_{a,b} \neq \emptyset$, which is Proposition 9.1(1). For the upper bound, $|\Sigma_{a,b}| = g$ would mean that every prime above 2 divides $\Lambda_{a,b}$, i.e. that $\Lambda_{a,b} \equiv 0 \pmod{2\mathbb{Z}[\zeta_{p^j}]}$, contradicting Theorem 11.3. Finally $g = \varphi(p^j)/f = p^{j-1}(p-1)/f \geq p^{j-1}$ because $f \mid p-1$ on the plateau. ■

This is the precise answer to the question in the title of this section. *The reciprocal, product-one symmetry of the five-term relation supplies at most a factor of 4, and generically a factor of 2, toward the $g \geq p^{j-1}$ primes above 2 that a Dvornicich–Zannier style argument needs.* The deficit is not a constant that could be absorbed by sharpening the inequality $\sum_{q|Q} (q-2) \leq k-2$; it grows like p^{j-1} .

There is also an archimedean constraint, which we record because it shows how far the “all primes above 2” case is from being tight. By Proposition 9.1(3)–(4), a counterexample forces $2^{f|\Sigma|} \mid \mathrm{Res}(\Phi_{p^j}, G_r)$, while

$$\frac{1}{\varphi(p^j)} \log |\mathrm{Res}(\Phi_{p^j}, G_r)| \longrightarrow m(G_r),$$

m denoting the logarithmic Mahler measure, and by Lawton’s theorem [Law83]

$$\lim_{r \rightarrow \infty} m(G_r) = m(1 + x + x^{-1} + y + y^{-1}) = 0.2513\dots = 0.3626\dots \times \log 2.$$

(The numerical value was computed by midpoint quadrature on a 4096×4096 grid; the polynomial belongs to Boyd’s family $x + x^{-1} + y + y^{-1} + k$ at $k = 1$ [Boy98].) Therefore, asymptotically, $f|\Sigma| \lesssim 0.3626 \varphi(p^j)$, i.e.

$$|\Sigma_{a,b}| \lesssim 0.3626 g.$$

Here the roots-of-unity limit and the Lawton limit are coupled through the choice of the residue class $r \bmod p^j$, so this estimate is heuristic rather than uniform in r . Thus the archimedean size heuristic is consistent with Theorem 11.3, while leaving the whole range $2 \leq |\Sigma| \lesssim 0.36 g$ open.

12 The cyclic-code and Zetterberg-code attack

Fix an odd n and an element θ of exact order n . The local vanishing code is

$$\mathcal{K}_n = \left\{ (c_t)_{t \in \mathbb{Z}/n\mathbb{Z}} \in \mathbb{F}_2^n : \sum_t c_t \theta^t = 0 \right\}.$$

It is a binary cyclic code of dimension $n - \text{ord}_n(2)$. A nondegenerate five-term relation is the codeword with support $\{0, \pm a, \pm b\}$. This is a *single-prime* code, unlike the ideal $\Phi_{p^j}R$ in Theorem 11.3, which imposes vanishing at every prime above two.

Theorem 12.1 (Interleaving off the plateau). *Let $n = p^j$ with $j > c_p$, let θ have order p^j , and put $\eta = \theta^p$. Every polynomial of degree less than p^j has a unique decomposition*

$$A(X) = \sum_{r=0}^{p-1} X^r A_r(X^p), \quad \deg A_r < p^{j-1}.$$

Then

$$A(\theta) = 0 \iff A_r(\eta) = 0 \text{ for every } r.$$

Thus $\mathcal{K}_{p^j}$ is the interleaving of p copies of $\mathcal{K}_{p^{j-1}}$. Every codeword of weight at most five is supported in one residue class modulo p .

Proof. For $j > c_p$ one has

$$[\mathbb{F}_2(\theta) : \mathbb{F}_2(\eta)] = \frac{\text{ord}_{p^j}(2)}{\text{ord}_{p^{j-1}}(2)} = p.$$

Hence $1, \theta, \dots, \theta^{p-1}$ are linearly independent over $\mathbb{F}_2(\eta)$, and the equivalence follows. A nonzero component $A_r(\eta)$ cannot have weight one or two: $\eta^s \neq 0$, and $\eta^s + \eta^t = 0$ forces $s \equiv t \pmod{p^{j-1}}$. If two interleaved components were nonzero, the total weight would therefore be at least six. ■

Corollary 12.2. *The interleaving theorem gives another proof of the five-term descent for $j > c_p$: since the support contains the exponent 0, all five exponents lie in the zero residue class modulo p , so $p \mid a, b$.*

On the plateau the extension degree in the proof is one, so the interleaving collapses. In the nonsplit case, with $f = 2\rho_2(p^j)$ and $q = 2^{\rho_2(p^j)}$, the group μ_{p^j} lies in the norm-one circle of order $q+1$ in $\mathbb{F}_{q^2}$. The code $\mathcal{K}_{p^j}$ is then a shortening of the binary Zetterberg code on that circle. The standard minimum-distance result for the ambient code is five [LW90]; shortening rules out weights below five, but the forbidden relation has weight exactly five. Thus the general code bound reaches the boundary and stops.

Weight enumerators or classifications of minimum Zetterberg words do not finish the problem either. They classify words on the full norm-one circle, whereas the endpoint asks whether a minimum word has the special reciprocal support

$$\{1, \theta^{\pm a}, \theta^{\pm b}\}$$

inside one prescribed p -Sylow subgroup. Retaining that support condition turns the classification back into the original intersection problem. In split cases the analogous irreducible cyclic-code bounds are weaker still. Coding theory therefore gives a clean structural proof off the plateau and an exact explanation of why minimum-distance technology alone cannot cross it.

13 Counterexamples in other residue characteristics

Every structural ingredient used so far — the cyclotomic field $\mathbb{Q}(\zeta_{p^j})$, its Galois group, the decomposition group $\langle \ell \rangle$ at a prime above ℓ , the classification of exact vanishing sums, the Dvornicich–Zannier inequality,

the reciprocal product-one symmetry, and the plateau condition $c_p \geq j$ — makes sense verbatim with 2 replaced by an arbitrary prime ℓ . We now show that the resulting statement is false.

For a prime $\ell \neq p$ define

$$f_1(\ell, p) = \text{ord}_p(\ell), \quad c(\ell, p) = v_p(\ell^{f_1} - 1),$$

so that $c(\ell, p) \geq 2$ says exactly that p is a Wieferich prime to base ℓ , and $\mu_{p^j} \subseteq \mathbb{F}_{\ell^{f_1}}$ for $j \leq c(\ell, p)$.

Theorem 13.1. *The analogue of Problem 1.1 with 2 replaced by ℓ has a negative answer. Each row of Table 1 gives a prime ℓ , a prime $p > 5$, an exponent $j \geq 2$, an element θ of exact order p^j in $\mathbb{F}_\ell$, and integers a, b prime to p with*

$$1 + \theta^a + \theta^{-a} + \theta^b + \theta^{-b} \equiv 0 \pmod{\ell},$$

and with $1, \theta^{\pm a}, \theta^{\pm b}$ pairwise distinct. In every row $c(\ell, p) \geq j$, so the example lies on the base- ℓ plateau.

ℓ	p	j	p^j	θ	a	b	$c(\ell, p)$
491	7	2	49	42	9	15	2
1373	7	3	343	16	29	152	3
29009	7	2	49	7394	4	19	2
727	11	2	121	358	2	30	2
677	13	2	169	16	28	40	2

Table 1: Verified counterexamples over prime fields. Each is checkable by five modular exponentiations.

Proof. Each row is a finite verification, carried out in `code/ai_prototypes/plateau_scan.py` and independently rechecked with plain modular arithmetic. For instance, in $\mathbb{F}_{491}$ the element $\theta = 42$ satisfies $42^{49} \equiv 1$ and $42^7 \not\equiv 1$, and

$$1 + 42^9 + 42^{40} + 42^{15} + 42^{34} \equiv 0 \pmod{491},$$

with $\gcd(9, 7) = \gcd(15, 7) = 1$. Here $491 \equiv 1 \pmod{49}$, so $f_1 = 1$ and $c(491, 7) = v_7(490) = 2$. ■

Further counterexamples occur over non-prime fields, for example $\ell = 97$, $p = 7$, $j = 2$ in $\mathbb{F}_{97}[x]/(x^2 + 18x + 1)$ with $(a, b) = (11, 17)$; $\ell = 241$, $p = 11$, $j = 2$ with $(a, b) = (17, 18)$; $\ell = 337$, $p = 13$, $j = 2$ with $(a, b) = (9, 24)$; $\ell = 577$, $p = 17$, $j = 2$ with $(a, b) = (21, 38)$; and $\ell = 41$, $p = 29$, $j = 2$ in a quartic extension with $(a, b) = (4, 7)$.

Corollary 13.2. *No proof of Problem 1.1 can use only the following inputs: the arithmetic of $\mathbb{Q}(\zeta_{p^j})$ and its decomposition groups; the classification of exact vanishing sums of roots of unity (Mann [Man65], Conway–Jones [CJ76], Lenstra [HWL79], Lam–Leung [LL00]); congruence analogues of the Conway–Jones inequality [DZ02] and its number-field-coefficient companion [DZ00]; the reciprocal product-one symmetry of the five terms; and the plateau condition. All of these hold verbatim in the configurations of Table 1.*

Remark 13.3 (a characteristic-three accident). *The smallest plateau configuration of all, $\ell = 3$ and $p = 11$ (with $3^5 - 1 = 242 = 2 \cdot 11^2$, so $c(3, 11) = 2$ and $\mu_{121} \subseteq \mathbb{F}_{243}$), has no solutions, even though the naive count $p^{2j}/\ell^{f_1} = 121^2/243 \approx 60$ predicts many. The reason is special to characteristic three. In $\mathbb{F}_{243}^\times$ the group μ_{121} has index two, hence is the group of squares, so every $u \in \mu_{121}$ is w^2 and*

$$u + u^{-1} = (w + w^{-1})^2 - 2 = s^2 + 1, \quad s = w + w^{-1}.$$

Writing $v = z^2$ and $t = z + z^{-1}$ likewise, the equation $1 + u + u^{-1} + v + v^{-1} = 0$ becomes $3 + s^2 + t^2 = 0$, that is $s^2 + t^2 = 0$. If $t = 0$ then $v + v^{-1} = 1$, so $v^2 - v + 1 = (v + 1)^2 = 0$ in characteristic three and $v = -1$, which does not lie in the odd-order group μ_{121} ; so $t \neq 0$ and $(s/t)^2 = -1$. But -1 has order two in $\mathbb{F}_{243}^\times$, a group of order $242 = 2 \cdot 11^2$, whereas every square has odd order; so -1 is not a square and there is no solution. This is precisely the kind of characteristic-specific accident that a proof for $\ell = 2$ would have to exhibit, and it has no analogue at $\ell = 2$.

14 The quantitative criterion

Table 1 and the scan of `code/ai_prototypes/plateau_scan.py` are explained by a single heuristic.

Heuristic 14.1. *Let $n = p^j$ and*

$$\rho_\ell(n) = \min\{r > 0 : \ell^r \equiv \pm 1 \pmod{n}\}, \quad Q = \ell^{\rho_\ell(n)}.$$

The number of pairs $(u, v) \in \mu_n^2$ with $1 + u + u^{-1} + v + v^{-1} = 0$ is about n^2/Q . Solutions are therefore expected exactly when

$$p^j \gtrsim \ell^{\rho_\ell(p^j)/2}. \quad (26)$$

Equivalently, the p -power subgroup must be at least the square root of the split or norm-one torus containing it.

The heuristic is simply that $\Gamma = \{v + v^{-1} : v \in \mu_n\}$, a set of $(n+1)/2$ elements contained in $\mathbb{F}_Q$, should meet the translate $-1 - \Gamma$ in about $|\Gamma|^2/Q$ points. Each nondegenerate trace intersection lifts to four ordered pairs (u, v) , producing the estimate n^2/Q . By Lemma 1.7, Γ is the root set of $sF_n(s)$. It matches the data: over $\mathbb{F}_{491}$ the ratio $49^2/491 \approx 4.9$ predicts a handful of solutions and there are 8. Over $\mathbb{F}_{9409} = \mathbb{F}_{97^2}$ the group μ_{49} is nonsplit and its trace values lie in $\mathbb{F}_{97}$, so the relevant ratio is $49^2/97 \approx 24.8$, consistent with the observed 32 solutions. Over $\mathbb{F}_{2^{21}}$ (the case $p = 7, j = 2, \ell = 2$) the ratio is $49^2/2^{21} \approx 0.001$ and there are none.

On the base-2 plateau, put $\rho = \rho_2(p)$. The group μ_{p^c} lies in a torus of size $2^\rho - 1$ or $2^\rho + 1$, while (26) requires $p^c \gtrsim 2^{\rho/2}$. This isolates the approximate window

$$c \log_2 p + O(p^{-c}) \leq \rho_2(p) \lesssim 2c \log_2 p. \quad (27)$$

Equivalently, $2^\rho - 1$ in the split case or $2^\rho + 1$ in the nonsplit case would need a plateau prime-power divisor at least as large as its square root.

p	c_p	$\rho_2(p)$	window (27)	$p^{2c}/2^{\rho_2(p)}$	margin
1093	2	182	(20.2, 40.4]	$10^{-42.6}$	$4.5 \times$
3511	2	1755	(23.6, 47.1]	$10^{-514.1}$	$37.3 \times$

Table 2: The two known base-2 Wieferich primes miss the counterexample window (27) by the stated factor.

Remark 14.2. *A prime p is Wieferich if and only if $p^2 \mid 2^{\text{ord}_p(2)} - 1$. Hence the conjecture that every Mersenne number $2^n - 1$ is squarefree would imply $c_p = 1$ for all p and, by Corollary 3.6, would settle Problem 1.1 outright. That conjecture is false, since $1093^2 \mid 2^{364} - 1$. The heuristic asks for a different, still inaccessible statement: no plateau prime power p^c dividing the relevant factor $2^{\rho_2(p)} \pm 1$ should be as large as its square root. We are not aware of a technique that proves such a bound.*

Remark 14.3 (why character sums cannot help). *Writing ψ for the canonical additive character of $\mathbb{F}_q$ and $S(t) = \sum_{u \in \mu_n} \psi(t(u + u^{-1}))$, the solution count is $N = q^{-1} \sum_t \psi(t) S(t)^2$. Each $S(t)$ with $t \neq 0$ is an average of $(q-1)/n$ twisted Kloosterman sums, so $|S(t)| \leq 2\sqrt{q}$ and $\sum_t |S(t)|^2 = q(2n-1)$. These give only $N \leq 2n-1$, which is the trivial bound. Even an optimal square-root cancellation estimate has error term $\gg \sqrt{q}$, which exceeds the main term n^2/q whenever $n^2 < q^{3/2}$. For $p = 1093$ and $j = 2$ this reads $1093^4 \approx 10^{12}$ against $2^{546} \approx 10^{164}$. No equidistribution statement can therefore prove $N = 0$; only an exact algebraic identity can.*

15 Computations

All computations are exact except for two clearly marked places: the numerical Mahler measure quoted in Section 11, and the complex-arithmetic sweep used to corroborate Theorem 7.1. Every assertion about a finite field, a resultant, or a certificate is exact integer arithmetic.

code/ai_prototypes/five_term_relations.py Characteristic-agnostic solver. Builds $\mathbb{F}_{\ell^f}$ from a Rabin-tested irreducible polynomial, locates an element of exact order p^j , tabulates $\Gamma = \{\theta^b + \theta^{-b}\}$ and intersects it with $-1 - \Gamma$. Validated on $(\ell, p, j) = (2, 5, 1)$, where it recovers the eight solutions coming from $\mathcal{R}_5$.

code/ai_prototypes/plateau_scan.py Enumerates all triples (ℓ, p, j) with $p > 5$, $j \geq 2$, $c(\ell, p) \geq j$ and ℓ^{f_1} below a bound, runs the solver on each, and emits and rechecks a certificate consisting of the minimal polynomial of θ over $\mathbb{F}_\ell$ together with (a, b) . The scan over all bases $\ell \leq 1000$ with $\ell^{f_1} \leq 2 \times 10^7$ produced the counterexamples cited in Section 13.

code/ai_prototypes/five_term_char2.py Dedicated characteristic-two solver. Elements of $\mathbb{F}_{2^f}$ are Python integers used as $\mathbb{F}_2$ -coefficient bit vectors; the field is re-coordinatized so that $\theta = x$, making each successive power a shift and a conditional XOR. Membership is decided on 64-bit fingerprints — legitimate because addition is XOR, so the fingerprint of a sum is the XOR of fingerprints — and every fingerprint collision is re-verified exactly. For the Wieferich prime $p = 1093$ and $j = 2$, with $n = 1194649$ and $f = \text{ord}_{1093^2}(2) = 364$, the search completes in under two seconds and finds *no* fingerprint candidates at all, hence no solutions. For $p = 3511$ and $j = 2$, with $n = 12327121$ and $f = \text{ord}_{3511^2}(2) = 1755$, the search completes in under four minutes and again finds no candidates. Since $c_{1093} = c_{3511} = 2$, these are the only plateau exponents for the two known Wieferich primes, so together with Theorem 3.5 they settle Problem 1.1 for both. This is also an independent confirmation, by a method that never forms a Fibonacci polynomial, of $d(1093^2 - 1) = d(3511^2 - 1) = 0$.

code/ai_prototypes/five_term_resultants.py Computes, for a given p^j , the complete table of relative norms $N_{K^+/\mathbb{Q}}(\Lambda_{1,r})$ over all classes of nonzero ratios r , together with their factorizations, and reports whether 2 occurs. This is the exact criterion of Proposition 9.1; running it on $(7, 2)$ reproduces Table 3.

code/ai_prototypes/verify_p7_characteristics.py Checks the prediction of Table 3: for each prime ℓ dividing one of the fourteen relative norms, the five-term relation does have a solution over $\overline{\mathbb{F}}_\ell$ with u or v of exact order 49. All eight predicted characteristics were confirmed.

code/ai_prototypes/sine_unit_obstruction.py The computations of Section 10. It tabulates the sine residues $\bar{\lambda}_x$, verifies the fibre structure of Lemma 10.2, enumerates all product-form collisions and their ratio classes (confirming Theorem 10.9 for every $p \leq 257$ with $\text{ord}_p(2) \leq 60$), and computes the power-residue symbols Ψ_ℓ of Definition 10.13. For the p -primary symbol it checks Theorem 10.15 by comparing the Fourier support of Ψ against the admissible eigencharacters A ; over all primes $7 \leq p \leq 199$ with $\text{ord}_p(2) \leq 200$ the support was contained in A in every case, Ψ vanished identically exactly when $\text{ord}_p(2)$ was even, and the ratio classes admitted by (24) with $\ell = p$ always contained the ratio classes of the actual solutions. The obstruction was empty — hence a proof of $d(p - 1) = 0$ — for $p = 7, 23, 47, 71, 79, 89, 103, 167, 191, 199$, and exactly sharp for $p = 31$ and $p = 127$. The same script verifies the eigenrelation of Theorem 10.15(1) and the descent of Theorem 10.15(3) at levels $j = 2, 3$, verifies Corollary 10.16 for $p = 1093$, $j = 2$ inside $\mathbb{F}_{2^{364}}$, and runs the test over all primes $\ell \mid 2^f - 1$ of Lemma 10.14: for $p = 17$ and $p = 31$, where solutions exist, every ℓ admits pairs, while for $p = 23$ and for $(p, j) = (7, 2)$ every ℓ excludes them.

code/ai_prototypes/plateau_ppart_3511.py The plateau test of Example 10.29. It splits Φ_{3511} over $\mathbb{F}_2$ by a trace map to obtain $\mathbb{F}_{2^{1755}}$, locates θ of exact order 3511^2 , confirms the eigenrelation, and computes Ψ on a sample of the group G_p of principal units by baby-step giant-step in μ_{p^2} . With 140 samples it exhibits 4 pairs satisfying the p -primary necessary condition, against 2.79 expected at random: the obstruction that settles $j = 1$ does not settle $j = 2$.

code/cn_experiments/verify_all.py Verifies the shifted-Dickson recurrence and power-sum formula, the Hasse-derivative generating function, the Multiplicity-Two identity $C_n^{[2]}(Y) = Y^{-2}$, and the bridge (11). The supporting scripts **code/cn_experiments/integer_verify.py** and **code/cn_experiments/factorization_norms.py** verify the integer lift, telescoping identity, real-cyclotomic factorization, and discriminant/resultant certificates.

`code/cn_experiments/collision_parity.py` Checks the collision involution, the internal and mixed resultant formulas, and the plateau inversion criterion. `code/cn_experiments/relative_norm_identity.py` verifies (20) and its resultant descent, while `code/cn_experiments/dynatomic_orbits.py` verifies the covering-degree bound and the doubling-orbit factorization of Theorem 5.4.

`code/ai_prototypes/circular_module.py` The computations of Sections 10.2, 10.6, and 10.8. It verifies Theorem 10.21 for every index and every $p \leq 257$ with $\text{ord}_p(2) \leq 40$ and for $(p, j) = (7, 2), (23, 2)$; verifies the level pattern of Theorem 10.22 and its two proof ingredients for $p = 7, 23, 47, 71, 79, 103, 167, 191, 199$; confirms the group-ring structure of the semi-local residue module by Smith normal form; checks the freeness of the admissible-system parameter space against the true system; constructs explicit injective phantom systems with primitive collisions for $(1093, 2)$ and $(3511, 2)$; and verifies the Kloosterman nullity formulas against direct enumeration for all Mersenne primes up to 131071 and all Fermat primes up to 65537.

`code/ai_prototypes/plateau_endpoint.py` Verifies the endpoint Sylow reduction, the box-plus-centre form, the global trace law (25), the cases $p = 73, 89, 233$, the elliptic model of Proposition 2.8, the additive distribution law, and the strengthened phantom systems at $(1093, 2)$ and $(3511, 2)$.

`code/ai_prototypes/sp4_explicit.py` Checks the symplectic form, characteristic polynomial, and element order in the explicit level-one Sp_4 examples used to test the torus classification.

15.1 The complete resultant table for $p^j = 49$

By Proposition 9.1 the whole of Problem 1.1 for a given (p, j) is the assertion that finitely many explicit integers are odd. For $p^j = 49$ the ratio $r = b/a$ ranges over the nonzero residues modulo 49, and the relative norm $N_{K^+/\mathbb{Q}}(\Lambda_{1,r})$ depends only on the class $\{\pm r^{\pm 1}\}$ (only $\{\pm r\}$ when r is not a unit). There are eleven unit classes and three non-unit classes. Table 3 lists them all.

class	$N_{K^+/\mathbb{Q}}(\Lambda_{1,r})$	factorization
$\{1, 48\}$	13555459	$97 \cdot 139747$
$\{2, 24, 25, 47\}$	1	unit (Corollary 8.2)
$\{3, 16, 33, 46\}$	1	unit (Corollary 8.2)
$\{4, 12, 37, 45\}$	3527	prime
$\{5, 10, 39, 44\}$	1373	prime
$\{6, 8, 41, 43\}$	97	prime
$\{9, 11, 38, 40\}$	881	prime
$\{13, 15, 34, 36\}$	1	unit (sporadic)
$\{17, 23, 26, 32\}$	29009	prime
$\{18, 19, 30, 31\}$	491	prime
$\{20, 22, 27, 29\}$	97	prime
$\{7, 42\}$	97	prime
$\{14, 35\}$	1	unit
$\{21, 28\}$	197	prime

Table 3: The fourteen relative norms for $p^j = 49$; the last three rows are the classes with $7 \mid r$, i.e. with θ^b of order dividing 7. Every entry is odd, which is exactly the assertion of Problem 1.1 for $p = 7, j = 2$.

Corollary 15.1. *For $p = 7$ and $j = 2$, the residue characteristics ℓ in which (4) has a solution with θ^a or θ^b of exact order 49 are exactly*

$$97, \quad 197, \quad 491, \quad 881, \quad 1373, \quad 3527, \quad 29009, \quad 139747,$$

and 2 is not among them.

Proof. By Proposition 9.1(1) and (4), a prime ℓ admits a solution with ratio r if and only if ℓ divides $\text{Res}(\Phi_{49}, G_r)$. Table 3 lists the complete factorizations. The verification that each of these primes really does admit a solution is the content of `code/ai_prototypes/verify_p7_characteristics.py` and of the scan in `code/ai_prototypes/plateau_scan.py`. The characteristics $\ell = 139747$ and $\ell = 197$ are degenerate: the first arises only from the class $r = \pm 1$, where $v = u^{\pm 1}$, and the second only from the class $r = \pm 21$, where $7 \mid b$. ■

Corollary 15.1 is the sharpest available description of the situation. For $p = 7$, $j = 2$ the truth of Problem 1.1 is not a structural fact; it is the statement that seven specific primes, produced by the factorizations of eleven cyclotomic resultants, happen not to include 2.

16 Summary

1. The shifted-Dickson sequence C_n is the Newton power-sum sequence of Q_S . Its repeated roots are exactly the dividing reciprocal quartics, every such root has multiplicity two, and $d(n-1) = 4 \deg B_n$ (Theorems 2.6 and 2.7). This removes all ambiguity about multiplicities in the polynomial formulation.
2. The degree argument proves $\text{ord}(\theta^a), \text{ord}(\theta^b) \leq p^{c_p}$. The universal theorem is therefore equivalent to the one endpoint statement

$$B_{p^{c_p}} = B_p,$$

or equivalently to the gcd in (14) being one (Theorem 3.8). All primes below 2^{64} satisfy the theorem.

3. The integer lift turns one stabilization step into the oddness of a discriminant and a resultant. Real-cyclotomic factorization shows that these are two independent obstructions: an internal primitive collision norm and a primitive-lower mixed norm (Theorem 4.5). The mixed norm is automatically odd off the plateau, but neither parity is forced by cyclotomic algebra on the plateau.
4. Arithmetic dynamics makes D_p a critically fixed map semiconjugate to $u \mapsto u^p$ through $u \mapsto u + u^{-1}$. It reproves mixed descent from the covering degree and factors B_{p^j} by doubling orbits (Theorems 5.3 and 5.4). On the plateau the degree does not grow, so the bound is vacuous and monodromy leaves the same finite collision-orbit problem.
5. The elliptic model identifies a relation with a point P on $Y^2 + XY = X^3 + X$ such that both $X(P)$ and $X(P + T)$ lie in μ_{p^j} , where T has order four (Proposition 2.8). The condition is on coordinates, not on a subgroup of the elliptic curve, so isogenies and point counts do not imply endpoint avoidance.
6. The Sp_4 model identifies a counterexample with a regular semisimple element of order p^j and trace one. Maximal-torus theory restricts it to two like SL_2 factors or a cyclic torus of order $q^2 \pm 1$, but p^j -torsion is abundant on the plateau and the trace-one condition is exactly the original incidence (Theorems 6.1 and 6.2).
7. Characteristic zero has only the order patterns $(5, 5), (3, 4), (2, 6)$, so for $p > 5$ the relevant algebraic integer is nonzero. Cyclotomic units eliminate exactly the structural ratio classes $\{\pm 2^{\pm 1}, \pm 3^{\pm 1}\}$ and no general additional class (Theorem 8.1).
8. Reduction modulo the rational ideal (2) is impossible by the Dvornicich–Zannier theorem or the elementary weight argument, but the problem concerns one prime above two. Reciprocal symmetry reaches at most four of the $g \geq p^{j-1}$ primes that an all-primes theorem needs (Theorem 11.6).
9. The sharp local statement is the Hilbert–90 congruence $s_{j,a} s_{j,b} \equiv 1$ at one prime above two. The semi-local target is $\mathbb{Z}[\widehat{G}]/(\sigma_2 - 2)$, while the Frobenius, sign, distribution, and norm axioms leave a free parameter space $(E^\times)^{g^+}$ (Theorems 10.5 and 10.7). Explicit phantom systems prove that those axioms do not imply collision avoidance.

10. Power-residue symbols and Stickelberger eigenspaces give strong necessary conditions and settle several level-one families. Their descent becomes the identity $0 = 0$ on a Wieferich plateau (Proposition 10.27); this failure is structural, not a defect of the calculation.
11. Additive trace identities go farther. The Artin–Schreier law proves the index-two family, including $d(3510) = 0$, and the global trace law proves $d(72) = d(88) = d(232) = 0$ (Theorems 10.21, 10.22, and 10.24). At the full-torus extremes the count is an exact binary Kloosterman sum (Theorem 10.26). The known linear trace constraints still admit formal plateau collisions, so a general proof must use more of the nonlinear addition law.
12. Coding theory is decisive off the plateau: the local cyclic code is an interleaving of lower-level codes, forcing every word of weight at most five to descend (Theorem 12.1). On a nonsplit plateau it is a shortened Zetterberg code, whose general minimum-distance theory stops at weight five, exactly the forbidden weight.
13. The theorem is false after replacing characteristic two by other residue characteristics (Theorem 13.1). Hence characteristic-zero classification, cyclotomic Galois structure, torus classification, and base-independent monodromy cannot suffice. A proof must exploit a special additive feature of characteristic two.
14. The size heuristic predicts a counterexample only when $\rho_2(p)$ lies in the narrow window (27). The known Wieferich primes 1093 and 3511 lie far outside it, and exact searches verify their endpoints. The complete $p^j = 49$ resultant table likewise contains no factor two.

The minimal unresolved assertion is therefore precise:

$$\boxed{\text{For every } p > 5 \text{ with } c_p \geq 2, \quad B_{p^{c_p}} = B_p.}$$

Equivalently, every endpoint collision in $\mu_{p^{c_p}}$ must already lie in μ_p , or the gcd in (14) must be one. Proving this single statement proves $d(p^k - 1) = d(p - 1)$ for every prime p and every k . The accumulated attacks show that the missing input must be both characteristic-two and subgroup-sensitive; none of the currently isolated degree, norm, trace, code, dynamics, or group-theoretic invariants alone has that strength.

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